

Thue–Morse Diagonal Laws

Diagonal Polynomials and Dyadic Block Geometry in Repeated Thue–Morse Summation

A consolidated volume for the Fabius-function corpus

Consolidated from three research reports with detailed proofs

September 5, 2026

Abstract

Consider the two-dimensional table specified by the signed Thue–Morse prefix row and the weighted recurrence

$$s_{n,k} = \sum_{j=0}^{k-1} (k-j) s_{n-1,j}.$$

Its first visible diagonals are polynomial in the row index, and the originating program listed eight such laws. This volume derives the polynomial on every diagonal and everything the three reports it consolidates could say about it. Row n is the $(2n+1)$ -fold inclusive prefix sum of the signed sequence shifted by $n+1$ columns; with $r = k - n - 1 \geq 0$,

$$s_{n,n+1+r} = D_r(n), \quad D_r(x) = \sum_{q=0}^{\lfloor r/2 \rfloor} \varepsilon_q \binom{2x+r-2q-1}{r-2q}, \quad \sum_{r \geq 0} D_r(x) z^r = \frac{E(z^2)}{(1-z)^{2x}},$$

where $E(z) = \prod_{j \geq 0} (1 - z^{2^j}) = \sum \varepsilon_q z^q$. Thus D_r has exact degree r and leading coefficient $2^r/r!$, all its monomial coefficients are unsigned-Stirling sums, and the shifted table is the proper Riordan array $(E(z^2), z/(1-z)^2)$; a general subdiagonal theorem for proper Riordan arrays explains the polynomiality. The logarithm of the diagonal generating function has the 2-adic cumulants $\lambda_m(x) = 2x + 2 - 2^{\nu_2(m)+1}$, so the polynomials obey a Newton–Bell recurrence, partition and determinant formulas, a Sheffer half-step lowering law $D_r(x) - D_r(x - \frac{1}{2}) = D_{r-1}(x)$, translation, differential and addition laws, and a binary Mahler self-similarity. The arithmetic is exact: $r!D_r(x)/2^{\lceil r/2 \rceil}$ is a primitive integer polynomial, which gives the exact common denominator of every diagonal, and every rational zero is an integer or a half-integer. All nonnegative rational zeros are classified by a dyadic residue condition, negative half-grid values are finite backward differences of the signed sequence, and an infinite family of negative integer roots is proved.

For each fixed row the entire sequence is a signed repetition of one nonnegative block of length 2^{2n+1} ; the volume proves its support, circular reflection, complement law, exact central plateau, maximum, mass, moments, multiplicities and number of distinct values, and the finite-block theorem for every order of iterated summation from which these follow. The block coefficients are the level- $2n$ Rvachev histogram of the corpus, and suitably scaled rows converge to the up-function and the Fabius function. Exact algorithms are given for every regime: symbolic generation of any diagonal, direct evaluation near the edge, signed-block reduction, and a logarithmic-time evaluator through signed Thue–Morse prefix moments. The three exact verification programs and Wolfram Language implementations shipped with the sources are retained.

Relation to the atlas and to the three sources

The atlas [1] proves the exact prefix collapse, the iterated-prefix convolution and generating function, terminal dyadic zero runs, and signed reversal, with Lean crosswalks for those statements; the repository's Lean development also formalizes the finite approximation polynomials, the corrected normalization, and the step approximants [4, 5]. Those facts are inputs here. The volume's own contribution is the reorganization around the two indices (n, k) and the complete fixed-diagonal polynomial calculus.

The three sources. This volume replaces three independently written reports on the same table, filed together on September 3, 2026: a proof-oriented note organized around the diagonal coordinate, Sheffer and ruler recurrences, half-integer roots and the finite-block geometry of every summation order; a report emphasizing the Riordan-array structure, 2-adic Newton–Bell recurrences, primitive normalization and exact denominators, the rational-root restriction, and a fast prefix-moment evaluator; and a report emphasizing the pulse polynomial of every row, the exact content of doubled rising factorials, the diagonal Mahler equation, the half-grid transposition, a provable family of negative integer roots, and the exact identification of the rows with the Rvachev histogram approximants. All three proved the same core (the table identification, the row and bivariate generating functions, the sparse rising-factorial formula, the monomial coefficients, the half-step, differential and addition laws, the 2-adic recurrence, the nonnegative half-integer root criterion, the negative half-grid formula, and the finite-block factorization). Each such result is stated once, in the form that carried the most complete proof, and the two other reports' variant formulations are kept only where they add a formula. The first report's notation is used throughout; the provenance appendix records the dictionary. Statements imported verbatim from the second and third reports write the diagonal index as m and q respectively where the first writes r ; the polynomial is the same D .

The volume loads the corpus notation file `docs/fabius-notation.tex` for consistency with the other volumes. No Lean formalization is claimed for any result here; the closing chapter lists the formalization targets.

Contents

Abstract	i
Relation to the atlas and to the three sources	ii
1 Problem, conventions, and main answer	1
1.1 The table	1
1.2 The diagonal coordinate	1
1.3 Summary of the structural results	2
2 Signed Thue–Morse prefixes and formal products	4
2.1 The lacunary product	4
2.2 Exclusive and inclusive prefixes	4
3 Generating function of the complete two-dimensional table	6
3.1 The weighted recurrence as two summations	6
3.2 A local discrete equation	7
3.3 Two bivariate generating functions	7
4 Why fixed-index iterated sums are polynomials	8
4.1 A general summation-order polynomial	8
4.2 Mahler deflation and sparse diagonals	9
5 Riordan-array structure and a general subdiagonal theorem	10
5.1 The shifted table is a proper Riordan array	10
5.2 Why fixed subdiagonals are always polynomials	10
5.3 General Bell and Sheffer descriptions	11
5.4 Application to the present table	12
6 The diagonal polynomials	13
6.1 First examples and agreement with the supplied code	13
6.2 Degree, leading layers, and integrality	13
6.3 Fixed-diagonal asymptotics	15
6.4 Finite-product and inclusion–exclusion forms	15
6.5 A basis over integer-valued polynomials	15
7 Sheffer structure and efficient recurrences	17
7.1 A Sheffer family	17
7.2 Half-step lowering	17
7.3 Differentiation and addition	18
7.4 The ruler-function recurrence	18
7.5 Bell-polynomial form	19
7.6 Translation formulas on the half-grid	20
8 Exact denominators, primitive normalization, and rational zeros	21
8.1 A primitive integer normalization	21
8.2 The exact common denominator	22
8.3 Every rational zero lies on the half-integer lattice	22

9	Exact half-integer roots and factor patterns	23
9.1	Diagonal-prefix transposition	23
9.2	Nonnegative half-integers	23
9.3	Complete classification of nonnegative rational roots	25
9.4	Negative half-integers as finite differences	26
9.5	Strict negative half-integers and a conjecture	26
9.6	A provable family of negative integer roots	27
10	Finite dyadic blocks for arbitrary summation order	29
10.1	The block polynomial	29
10.2	The block as a probability mass function	31
11	Specialization to the rows of the table	32
11.1	One positive block and signed repetition	32
11.2	Complete geometry of the first block	32
11.3	Bounded-composition interpretation	34
11.4	Normalized profiles and the Fabius connection	34
11.5	The first nontrivial blocks and a table of the diagonal array	35
12	Connection with the Fabius and Rvachev approximation layer	36
12.1	Exact row-histogram identification and the table limits	37
13	Algorithms and Wolfram Language implementation	39
13.1	Replacing the finite list of diagonal rules	39
13.2	Generating consecutive polynomials by the ruler recurrence	40
13.3	Precomputing a complete row block	40
13.4	Useful additional acceleration rules	41
13.5	Block reduction in the diagonal coordinate	41
13.6	Logarithmic-time signed prefix moments	41
13.7	Memory and implementation notes	43
13.8	Root and denominator utilities	43
13.9	Independent consistency checks	44
14	Computational verification	45
14.1	The three retained verification programs	45
14.2	Exact Python verification	46
14.3	Selected continued factorizations	47
15	Further deductions and research directions	48
15.1	Polynomial interpolation in a discrete parameter	48
15.2	Root automata and residual factors	48
15.3	Other Mahler products	48
15.4	Scaling regimes beyond fixed diagonals	48
15.5	Residual factors and zero geometry	49
15.6	Automatic and p -adic structure	49
15.7	Binary classification of negative rational factors	49
15.8	The residual root geometry	50
15.9	Joint asymptotics beyond a fixed diagonal	50
15.10	Fast coefficient queries inside one pulse	50
15.11	Formalization targets	50
A	A compact theorem dictionary	52
B	Nonnegative rational factors through $m = 31$	53
C	Formula sheet for implementation	54

D	Source provenance, notation dictionary, and reproducibility	55
D.1	Sources and consolidation	55
D.2	Notation dictionary	56
D.3	Retained files	56
	Bibliography	57

Chapter 1

Problem, conventions, and main answer

1.1 The table

Let

$$\varepsilon_q = (-1)^{s_2(q)}, \quad q \in \mathbb{N}_0, \quad (1.1)$$

where $s_2(q)$ is the number of ones in the binary expansion of q . This is the signed Thue–Morse sequence. Throughout the volume, n and k are nonnegative integers. The originating program is read as defining $s_{n,k}$ with the triangular initial condition $s_{n,k} = 0$ for $k \leq n$, the signed-prefix row at $n = 0$, and the fallback recurrence

$$s_{n,k} = \sum_{j=0}^{k-1} (k-j)s_{n-1,j}, \quad n \geq 1. \quad (1.2)$$

The symmetry branches and the first eight diagonal formulas are computational shortcuts. We will derive them rather than assume them.

The repository’s Thue–Morse atlas proves the exact prefix collapse, the iterated-prefix convolution and generating function, terminal dyadic zero runs, and signed reversal, with explicit Lean crosswalks for those statements [1]. The present volume reorganizes those ingredients around the two indices (n, k) , and develops the fixed-diagonal polynomial calculus. No priority claim is made for identities that follow from standard formal power-series manipulations.

1.2 The diagonal coordinate

The first nonzero entry of row n occurs at $k = n + 1$. It is therefore natural to set

$$r = k - n - 1. \quad (1.3)$$

The visible diagonal $k - n = d$ in the original table corresponds to $r = d - 1$. We reserve $D_r(x)$ for the polynomial attached to this offset.

Closed form for an arbitrary diagonal. For every $r \geq 0$, define

$$D_r(x) = \sum_{q=0}^{\lfloor r/2 \rfloor} \varepsilon_q \binom{2x + r - 2q - 1}{r - 2q} = \sum_{q=0}^{\lfloor r/2 \rfloor} \varepsilon_q \frac{2x^{\overline{r-2q}}}{(r-2q)!}. \quad (1.4)$$

Then

$$s_{n,k} = \begin{cases} 0, & k \leq n, \\ D_{k-n-1}(n), & k \geq n+1. \end{cases} \quad (1.5)$$

Equivalently, the polynomial on the original diagonal $d = k - n \geq 1$ is

$$P_d(x) = D_{d-1}(x) = \sum_{q=0}^{\lfloor (d-1)/2 \rfloor} \varepsilon_q \binom{2x + d - 2q - 2}{d - 2q - 1}. \quad (1.6)$$

It has degree $d - 1$ and leading coefficient $2^{d-1}/(d-1)!$.

Formula (1.4) is finite and exact. It is not an interpolation guessed from numerical rows: it follows coefficientwise from a single generating function, and hence applies to every row and every nonnegative column.

Indexing guide. The row parameter n in the code is not the number of prefix summations. Row n contains the $(2n+1)$ -fold inclusive prefix sum, shifted right by $n+1$ columns. The natural diagonal coordinate is

$$m = k - n - 1.$$

The polynomial $D_m(x)$ describes the line $k = n + 1 + m$; its value at the integer $x = n$ is exactly the table entry on that line.

1.3 Summary of the structural results

Beyond the closed form, the volume proves the following.

1. The row generating function is

$$A_n(z) := \sum_{k \geq 0} s_{n,k} z^k = \frac{z^{n+1} E(z^2)}{(1-z)^{2n}} = \frac{z^{n+1} E(z)}{(1-z)^{2n+1}},$$

so the shifted row is exactly an odd-order iterated prefix, $s_{n,n+1+r} = T_r^{(2n+1)}$, and the shifted table is the proper Riordan array $(E(z^2), z/(1-z)^2)$ (Chapter 5). Every proper Riordan array has polynomial subdiagonals; here the degree is exactly r .

2. The sequence $\widehat{D}_r(x) = r! D_r(x)$ is Sheffer for $E(z^2) \exp(-2x \log(1-z))$. In particular

$$D_r(x) - D_r(x - \tfrac{1}{2}) = D_{r-1}(x), \quad r D_r(x) = \sum_{m=1}^r (2x + 2 - 2^{\nu_2(m)+1}) D_{r-m}(x),$$

with Bell, partition and determinant forms of the second identity, translation, differential and addition laws, and the binary self-similarity $G_x(z) = (1-z)(1+z)^{2x+1} G_x(z^2)$ of the generating function (Chapter 7).

3. $r! D_r(x)/2^{\lceil r/2 \rceil}$ is a primitive integer polynomial, the exact common denominator of D_r is $r!/2^{\min(\nu_2(r!), \lceil r/2 \rceil)}$, and every rational zero of D_r lies in $\frac{1}{2}\mathbb{Z}$ (Chapter 8).

4. For $m \geq 0$,

$$D_r(m/2) = 0 \iff r \bmod 2^{m+1} \in \{2^{m+1} - (m+1), \dots, 2^{m+1} - 1\},$$

and $D_r(m/2) = T_r^{(m+1)}$ reads a fixed column vertically through all prefix orders; negative half-integer values are finite backward differences of the signed sequence, and $D_r(-2^\ell) = 0$ whenever $r \equiv -1 \pmod{2^a}$ and ℓ is odd with $\ell \leq a - 1$ (Chapter 9).

5. A row has block length $L = 2^{2n+1}$, block maximum $M_n = 2^{\binom{2n}{2}}$, exactly $2n + 1$ zero positions, exactly $2n + 1$ maximum positions, and $2^{2n} - 2n + 1$ distinct absolute values; every nonextreme value occurs exactly twice, the block is palindromic and complement-symmetric, and its normalized coefficients are the probability mass function of a sum of independent discrete uniforms (Chapters 10 and 11). Divided by M_n they are the level- $2n$ Rvachev histogram samples, and the rows converge to the up-function and the Fabius function under the dyadic scaling (Chapter 12).
6. Every table entry can be computed exactly in $O(n^2 \log r)$ arithmetic operations, where $r = k - n - 1$, through a binary divide-and-conquer recurrence for the signed prefix moments $\sum_{q < N} \varepsilon_q q^d$ (Chapter 13).

Chapter 2

Signed Thue–Morse prefixes and formal products

2.1 The lacunary product

The binary recursions

$$\varepsilon_{2q} = \varepsilon_q, \quad \varepsilon_{2q+1} = -\varepsilon_q \quad (2.1)$$

imply the formal product identity

$$E(z) := \sum_{q \geq 0} \varepsilon_q z^q = \prod_{j \geq 0} (1 - z^{2^j}). \quad (2.2)$$

Every coefficient of the infinite product stabilizes after finitely many factors, so (2.2) is valid in $\mathbb{Z}[[z]]$; no analytic convergence is required. Separating the first factor gives the Mahler relation

$$E(z) = (1 - z)E(z^2). \quad (2.3)$$

These identities are standard in the theory of automatic sequences [9]; they are also the algebraic engine of the repository’s Fabius–Rvachev development [1].

2.2 Exclusive and inclusive prefixes

Define the exclusive prefix

$$S(k) = \sum_{j=0}^{k-1} \varepsilon_j, \quad S(0) = 0. \quad (2.4)$$

Pairing $\varepsilon_{2q} + \varepsilon_{2q+1} = 0$ gives

$$S(2q) = 0, \quad S(2q+1) = \varepsilon_q. \quad (2.5)$$

Consequently,

$$\sum_{k \geq 0} S(k) z^k = zE(z^2) = \frac{zE(z)}{1 - z}. \quad (2.6)$$

This is precisely the $n = 0$ row encoded by the supplied Wolfram Language branches.

For comparison with repeated summation, define inclusive iterates

$$T_r^{(0)} = \varepsilon_r, \quad T_r^{(h+1)} = \sum_{j=0}^r T_j^{(h)}. \quad (2.7)$$

One summation multiplies an ordinary generating function by $(1 - z)^{-1}$, so

$$\sum_{r \geq 0} T_r^{(h)} z^r = \frac{E(z)}{(1 - z)^h}. \quad (2.8)$$

For $h \geq 1$, coefficient extraction yields

$$T_r^{(h)} = \sum_{j=0}^r \binom{r - j + h - 1}{h - 1} \varepsilon_j. \quad (2.9)$$

Bertazzon's construction of a solution to a Thue–Morse integral equation uses these repeated sums and their dyadic block coefficients in essentially this form [11].

Chapter 3

Generating function of the complete two-dimensional table

3.1 The weighted recurrence as two summations

Let

$$A_n(z) = \sum_{k \geq 0} s_{n,k} z^k. \quad (3.1)$$

The kernel in (1.2) has generating function

$$\sum_{m \geq 1} m z^m = \frac{z}{(1-z)^2}. \quad (3.2)$$

Thus (1.2) is convolution by the coefficient sequence $1, 2, 3, \dots$, or equivalently two discrete antiderivatives together with a one-step shift.

Theorem 3.1 (Exact row generating function). *For every $n \geq 0$,*

$$A_n(z) = \left(\frac{z}{(1-z)^2} \right)^n z E(z^2) = \frac{z^{n+1} E(z^2)}{(1-z)^{2n}} = \frac{z^{n+1} E(z)}{(1-z)^{2n+1}}. \quad (3.3)$$

Proof. Equation (2.6) gives $A_0(z) = z E(z^2)$. Multiplying by (3.2) implements the recurrence once, so induction gives the first two expressions. The last follows from the Mahler identity (2.3). $\square$

The triangular zero region is now automatic: A_n is divisible by z^{n+1} . More importantly, shifting by this monomial identifies each row with a standard iterated-prefix sequence.

Corollary 3.2 (Rows are odd-order iterated prefixes). *For every $n, r \geq 0$,*

$$s_{n,n+1+r} = T_r^{(2n+1)}. \quad (3.4)$$

Equivalently,

$$s_{n,n+1+r} = \sum_{j=0}^r \varepsilon_j \binom{2n+r-j}{2n}. \quad (3.5)$$

Proof. After removing z^{n+1} from (3.3), the remaining generating function is $E(z)/(1-z)^{2n+1}$, which is (2.8) with $h = 2n+1$. $\square$

This relation also identifies the absolute values in the earlier question on periodic iterated Thue–Morse sums. With the notation used there,

$$|s_{n,n+1+r}| = u_r^{(2n)}. \quad (3.6)$$

The row index n therefore advances the summation order by two.

The first rows illustrate both the initial zero triangle and the rapid formation of a positive pulse, which [Chapter 11](#) describes exactly.

$n \backslash k$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	0	1	0	-1	0	-1	0	1	0	-1	0	1	0	1	0	-1	0
1	0	0	1	2	2	2	1	0	0	0	-1	-2	-2	-2	-1	0	0
2	0	0	0	1	4	9	16	24	32	40	48	55	60	63	64	64	64
3	0	0	0	0	1	6	20	50	104	190	316	490	719	1008	1360	1776	2256

3.2 A local discrete equation

The row generating functions form a geometric sequence, which turns the cumulative recurrence into a local second-difference law.

In particular,

$$\frac{A_n(z)}{A_{n-1}(z)} = \frac{z}{(1-z)^2} \quad (n \geq 1). \quad (3.7)$$

The cumulative recurrence in the original code is therefore the integrated form of a local second-difference law.

Corollary 3.3 (Local discrete equation). *With the convention $s_{n,k} = 0$ for $k < 0$, one has, for $n \geq 1$,*

$$s_{n,k} - 2s_{n,k-1} + s_{n,k-2} = s_{n-1,k-1}. \quad (3.8)$$

Proof. Multiply (3.7) by $(1-z)^2$ and compare coefficients of z^k . $\square$

Equation (3.8) fills a rectangular portion of the table in constant work per cell. It is often preferable to reevaluating the weighted sum in (1.2) for each new cell.

3.3 Two bivariate generating functions

Summing (3.3) over the row index gives a compact encoding of the whole array.

Proposition 3.4 (Bivariate table generating function). *As a formal power series in u with coefficients in $\mathbb{Z}[[z]]$,*

$$\sum_{n \geq 0} \sum_{k \geq 0} s_{n,k} u^n z^k = \frac{z(1-z)^2 E(z^2)}{(1-z)^2 - uz}. \quad (3.9)$$

For the diagonal array $D_r(n) = s_{n,n+1+r}$,

$$\sum_{n \geq 0} \sum_{r \geq 0} D_r(n) u^n z^r = \frac{(1-z)^2 E(z^2)}{(1-z)^2 - u}. \quad (3.10)$$

Proof. Both formulas are geometric sums. For example,

$$\sum_{n \geq 0} u^n A_n(z) = z E(z^2) \sum_{n \geq 0} \left(\frac{uz}{(1-z)^2} \right)^n.$$

Removing the shift z^{n+1} before summing gives (3.10). $\square$

Chapter 4

Why fixed-index iterated sums are polynomials

The polynomial phenomenon is not peculiar to Thue–Morse. It is a universal property of repeated summation, sharpened here by the Mahler factorization.

4.1 A general summation-order polynomial

Let $a_0, a_1, \dots$ be any sequence over a characteristic-zero field and let

$$A(z) = \sum_{j \geq 0} a_j z^j.$$

For an integer $h \geq 0$, write $U_r^{(h)}$ for the h -fold inclusive sum, so its generating function is $A(z)/(1-z)^h$.

Theorem 4.1 (Polynomiality in the summation order). *For each fixed $r \geq 0$, the coefficient*

$$U_r(h) := [z^r] \frac{A(z)}{(1-z)^h} \tag{4.1}$$

extends from $h \in \mathbb{N}_0$ to the polynomial

$$U_r(h) = \sum_{j=0}^r a_j \binom{h+r-j-1}{r-j}. \tag{4.2}$$

Its degree is at most r , and is exactly r if $a_0 \neq 0$, with leading coefficient $a_0/r!$.

Proof. Expand

$$(1-z)^{-h} = \sum_{m \geq 0} \binom{h+m-1}{m} z^m$$

formally and take the Cauchy coefficient at z^r . For fixed m , the generalized binomial coefficient is a polynomial in h of degree m . Only the term $j = 0$, for which $m = r$, can contribute degree r . $\square$

The value at $h = 0$ is included: the generalized binomial vanishes for positive lower index and upper index one smaller, leaving $U_r(0) = a_r$. This polynomial continuation is therefore canonical, not an arbitrary fit.

4.2 Mahler deflation and sparse diagonals

Many automatic sequences have a functional factorization that compresses the sum (4.2).

Theorem 4.2 (A general deflation principle). *Suppose*

$$A(z) = (1 - z)^c B(z^b), \quad B(z) = \sum_{q \geq 0} b_q z^q, \quad (4.3)$$

where $b \geq 2$ and $c \in \mathbb{Z}$. Then

$$U_r(h) = \sum_{q=0}^{\lfloor r/b \rfloor} b_q \binom{h - c + r - bq - 1}{r - bq}. \quad (4.4)$$

Thus only one residue class modulo b of coefficient degrees occurs.

Proof. Substitute (4.3) into (4.1):

$$\frac{A(z)}{(1 - z)^h} = B(z^b)(1 - z)^{-(h-c)}.$$

The term $b_q z^{bq}$ leaves coefficient degree $r - bq$ in the binomial series. □

For Thue–Morse, (2.3) has $b = 2$, $c = 1$, and $B = E$. Therefore

$$T_r^{(h)} = \sum_{q=0}^{\lfloor r/2 \rfloor} \varepsilon_q \binom{h + r - 2q - 2}{r - 2q}. \quad (4.5)$$

Since the table uses $h = 2n + 1$, formula (1.4) follows immediately. This proves the main answer (1.5).

Remark 4.3 (Interpretation). For a fixed coefficient index r , $D_r(x)$ is the polynomial continuation of the h -fold signed Thue–Morse prefix under the affine change $h = 2x + 1$:

$$D_r(x) = T_r^{(2x+1)} \quad \text{in the sense of polynomial continuation in } h. \quad (4.6)$$

At integer $x = n$, this is exactly the row identity (3.4).

Chapter 5

Riordan-array structure and a general subdiagonal theorem

5.1 The shifted table is a proper Riordan array

Shift the column index once and transpose the visual convention by setting

$$A_{r,n} = s_{n,r+1}, \quad r, n \geq 0. \quad (5.1)$$

Then $A_{r,n} = 0$ for $r < n$ and $A_{n,n} = 1$. From (3.3),

$$A_{r,n} = [z^r] E(z^2) \left(\frac{z}{(1-z)^2} \right)^n. \quad (5.2)$$

Thus A is the proper ordinary Riordan array

$$A = \left(E(z^2), \frac{z}{(1-z)^2} \right). \quad (5.3)$$

We use the convention that a Riordan pair (g, f) denotes the lower-triangular matrix $[z^r]g(z)f(z)^n$. Standard background can be found in [17, 18]; all facts needed below are also proved directly.

The compositional inverse of $f(z) = z/(1-z)^2$ is

$$\bar{f}(w) = \frac{1 + 2w - \sqrt{1 + 4w}}{2w} = \frac{\sqrt{1 + 4w} - 1}{\sqrt{1 + 4w} + 1} = w - 2w^2 + 5w^3 - 14w^4 + \dots. \quad (5.4)$$

Consequently the inverse unitriangular matrix is the Riordan array

$$A^{-1} = \left(\frac{1}{E(\bar{f}(z)^2)}, \bar{f}(z) \right). \quad (5.5)$$

This inverse is not needed to generate the table, but it shows that the table is an invertible change of basis over $\mathbb{Z}$: both f and $\bar{f}$ have integer coefficients and $E(0) = 1$.

5.2 Why fixed subdiagonals are always polynomials

The polynomial phenomenon is not accidental and is not special to Thue–Morse. It is a general feature of proper Riordan arrays.

Theorem 5.1 (Polynomial subdiagonals of a proper Riordan array). *Let K be a characteristic-zero field, let $g(z) \in K[[z]]$ satisfy $g(0) = g_0 \neq 0$, and let $\phi(z) \in K[[z]]$ satisfy $\phi(0) = 1$. Form the proper Riordan array*

$$R_{r,n} = [z^r] g(z) (z\phi(z))^n.$$

For every fixed $m \geq 0$, define the formal continuation of the m th subdiagonal by

$$P_m(x) = [z^m] g(z) \phi(z)^x, \quad \phi(z)^x := \exp(x \log \phi(z)). \quad (5.6)$$

Then:

1. $P_m(x) \in K[x]$ and $\deg P_m \leq m$;
2. $P_m(n) = R_{n+m,n}$ for every $n \in \mathbb{N}$;
3. if $\phi_1 = [z]\phi(z) \neq 0$, then $\deg P_m = m$ and

$$[x^m] P_m(x) = \frac{g_0 \phi_1^m}{m!}. \quad (5.7)$$

Proof. Write $L(z) = \log \phi(z)$. Since $\phi(0) = 1$, one has $L(z) \in zK[[z]]$. Therefore

$$g(z) \phi(z)^x = g(z) \sum_{j \geq 0} \frac{x^j L(z)^j}{j!}.$$

Because $L(z)^j$ has order at least j , only $j \leq m$ can contribute to $[z^m]$. This proves polynomiality and the degree bound. At a nonnegative integer $x = n$, the formal power $\phi(z)^x$ is the ordinary n th power, so

$$P_m(n) = [z^m] g(z) \phi(z)^n = [z^{n+m}] g(z) (z\phi(z))^n = R_{n+m,n}.$$

Finally, the coefficient of x^m can only come from $j = m$. To reach z^m in $g(z)L(z)^m$, one must take the constant g_0 and the linear term $\phi_1 z$ from every copy of L . This gives $g_0 \phi_1^m / m!$. $\square$

Remark 5.2. The theorem gives a canonical polynomial, not merely an interpolation through integer points. Formula (5.6) defines it in the coefficient ring before any values are sampled. This distinction matters when one studies coefficients, denominators, roots, or values at half-integer arguments.

5.3 General Bell and Sheffer descriptions

The same setup gives a general “shape theorem” finer than degree alone. Write

$$\log \left(\frac{g(z)}{g_0} \right) + x \log \phi(z) = \sum_{r \geq 1} \kappa_r(x) \frac{z^r}{r}, \quad (5.8)$$

where each $\kappa_r(x)$ is affine in x . This normalized formal logarithm is defined over K because $g(z)/g_0$ has constant term one. Differentiating the generating series in z gives the Newton recurrence

$$mP_m(x) = \sum_{r=1}^m \kappa_r(x) P_{m-r}(x), \quad P_0(x) = g_0. \quad (5.9)$$

Equivalently, $m!P_m(x)/g_0$ is a complete exponential Bell polynomial in $(r-1)!\kappa_r(x)$. Thus the lower coefficients of every Riordan subdiagonal are controlled by the logarithmic coefficients of g and ϕ .

Moreover, with $\tilde{D}_m(x) = m!P_m(x)$,

$$\sum_{m \geq 0} \tilde{D}_m(x) \frac{z^m}{m!} = g(z) e^{xf(z)}, \quad f(z) = \log \phi(z). \quad (5.10)$$

Hence $(\tilde{D}_m)$ is a Sheffer sequence. If $f'(0) \neq 0$ and $q = f^{(-1)}$ is the compositional inverse of f , the associated delta operator is $q(D_x)$ and satisfies

$$q(D_x) \tilde{D}_m(x) = mQ_{m-1}(x). \quad (5.11)$$

This is the finite-operator-calculus explanation of the especially simple half-step recurrence obtained later. See [14] for the general Sheffer formalism.

5.4 Application to the present table

For (5.3),

$$g(z) = E(z^2), \quad \phi(z) = \frac{1}{(1-z)^2}, \quad \phi_1 = 2.$$

The m th Riordan subdiagonal is

$$A_{n+m,n} = s_{n,n+m+1}. \quad (5.12)$$

This motivates the central definition.

The subdiagonal polynomial $P_m(x) = [z^m] E(z^2)(1-z)^{-2x}$ of (5.6) is exactly the diagonal polynomial $D_m(x)$ of (1.4), defined there by (7.2). [Theorem 5.1](#) therefore gives a second proof that D_m has exact degree m and leading coefficient $2^m/m!$, recovering (6.10), and the Riordan inverse (5.5) shows that the table is an invertible change of basis over $\mathbb{Z}$. Since $D_m(n)$ has degree m in n , its ordinary generating function in n is rational with denominator dividing $(1-u)^{m+1}$; the bivariate series (3.10) packages all of these rational functions at once. The general cumulants (5.8) specialize to the ruler cumulants of [Chapter 7](#), and the general lowering operator (5.11) to the half-step difference.

Chapter 6

The diagonal polynomials

6.1 First examples and agreement with the supplied code

Expanding (1.4) gives the following first eight members. They are exactly the eight explicit branches of the originating program after replacing x by n and r by $k - n - 1$.

$$D_0(x) = 1, \tag{6.1}$$

$$D_1(x) = 2x, \tag{6.2}$$

$$D_2(x) = (x + 1)(2x - 1), \tag{6.3}$$

$$D_3(x) = \frac{2}{3}x(x + 2)(2x - 1), \tag{6.4}$$

$$D_4(x) = \frac{1}{6}(4x^4 + 12x^3 - x^2 - 3x - 6), \tag{6.5}$$

$$D_5(x) = \frac{1}{15}x(x - 1)(4x^3 + 24x^2 + 39x + 34), \tag{6.6}$$

$$D_6(x) = \frac{1}{90}(x - 1)(x + 1)(2x - 1)(4x^3 + 32x^2 + 75x + 90), \tag{6.7}$$

$$D_7(x) = \frac{1}{315}x(x - 1)(x + 2)(2x - 1)(4x^3 + 40x^2 + 123x + 192). \tag{6.8}$$

The formula separates two kinds of structure. The rising-factorial terms control degree and coefficient growth, while the digital signs ε_q insert corrections only at degree gaps $2q$. This separation becomes transparent in the monomial coefficient formula below. The next polynomial, included to illustrate that the factor pattern is genuinely irregular, is

$$D_8(x) = \frac{x + 1}{2520} \left(\begin{array}{l} 16x^7 + 208x^6 + 856x^5 + 1384x^4 \\ - 1055x^3 - 3719x^2 + 3570x - 2520 \end{array} \right). \tag{6.9}$$

6.2 Degree, leading layers, and integrality

Theorem 6.1 (Basic shape). *For $r \geq 0$, the polynomial D_r has the following properties.*

1. $\deg D_r = r$ and

$$[x^r]D_r(x) = \frac{2^r}{r!}. \tag{6.10}$$

2. $r!D_r(x) \in \mathbb{Z}[x]$.

3. $D_r(x) \in \mathbb{Z}$ whenever $2x \in \mathbb{Z}$; in particular, every integer and half-integer value is integral.

4. At the origin,

$$D_r(0) = \begin{cases} \varepsilon_{r/2}, & r \text{ even}, \\ 0, & r \text{ odd}. \end{cases} \quad (6.11)$$

Thus every odd-indexed diagonal polynomial is divisible by x .

Proof. In (1.4), the $q = 0$ term has degree r and leading coefficient $2^r/r!$; every other term has degree at most $r - 2$. After multiplication by $r!$, each summand is an integer multiple of $2x^{r-2q}$, which belongs to $\mathbb{Z}[x]$. If $2x = t \in \mathbb{Z}$, then $\binom{t+m-1}{m} \in \mathbb{Z}$ for every $m \geq 0$, including negative t by the generalized-binomial identity. Finally, setting $x = 0$ kills every positive rising factorial, leaving only the term with $r - 2q = 0$, when one exists. $\square$

For even q , Vieta also gives the product of all roots:

$$\prod_{i=1}^q \rho_i = \varepsilon_{q/2} \frac{q!}{2^q}. \quad (6.12)$$

Let $\begin{bmatrix} m \\ \ell \end{bmatrix}$ denote the unsigned Stirling number of the first kind, characterized by

$$y^{\overline{m}} = \sum_{\ell=0}^m \begin{bmatrix} m \\ \ell \end{bmatrix} y^\ell. \quad (6.13)$$

Theorem 6.2 (All monomial coefficients). For $0 \leq \ell \leq r$,

$$\boxed{[x^\ell] D_r(x) = 2^\ell \sum_{q=0}^{\lfloor (r-\ell)/2 \rfloor} \varepsilon_q \frac{\begin{bmatrix} r-2q \\ \ell \end{bmatrix}}{(r-2q)!}.} \quad (6.14)$$

Consequently, the coefficient of x^{r-a} depends only on $\varepsilon_0, \dots, \varepsilon_{\lfloor a/2 \rfloor}$.

Proof. Insert (6.13) into the second form of (1.4) and compare powers of x . A term of degree $r - 2q$ can contribute to x^ℓ only if $q \leq (r - \ell)/2$. $\square$

The first three degree layers are especially simple. For $r \geq 3$,

$$\boxed{D_r(x) = \frac{2^r}{r!} x^r + \frac{2^{r-2}}{(r-2)!} x^{r-1} + \frac{2^{r-2}(3r^2 - 7r - 22)}{24(r-2)!} x^{r-2} + \mathcal{O}(x^{r-3}).} \quad (6.15)$$

For fixed r , this is also the large-row asymptotic shape of the diagonal entry $s_{n,n+1+r}$ as $n \rightarrow \infty$.

Derivation of (6.15). The leading term was already established. For degree $r - 1$, only $q = 0$ contributes, and $\begin{bmatrix} r \\ r-1 \end{bmatrix} = \binom{r}{2}$. For degree $r - 2$, use

$$\begin{bmatrix} r \\ r-2 \end{bmatrix} = \frac{r(r-1)(r-2)(3r-1)}{24}$$

and include the leading coefficient of the $q = 1$ term, whose sign is $\varepsilon_1 = -1$. Straight simplification gives the displayed coefficient. $\square$

The first three coefficients already determine useful global statistics of the roots.

Corollary 6.3 (First algebraic moments of the roots). *Let $\rho_1, \dots, \rho_q$ be the complex roots of D_q , counted with multiplicity, where $q \geq 2$. Then*

$$\sum_{i=1}^q \rho_i = -\frac{q(q-1)}{4}, \quad (6.16)$$

$$\sum_{1 \leq i < j \leq q} \rho_i \rho_j = \frac{q(q-1)(3q^2 - 7q - 22)}{96}, \quad (6.17)$$

$$\sum_{i=1}^q \rho_i^2 = \frac{q(q-1)(2q+11)}{24}. \quad (6.18)$$

Thus the arithmetic mean of all roots is $-(q-1)/4$.

Proof. Apply Vieta's formulas to [Equations \(6.10\) and \(6.15\)](#). The last identity follows from $(\sum_i \rho_i)^2 = \sum_i \rho_i^2 + 2 \sum_{i < j} \rho_i \rho_j$. $\square$

6.3 Fixed-diagonal asymptotics

Because q is fixed on one diagonal, the leading coefficients give a finite asymptotic expansion as $n \rightarrow \infty$:

$$D_q(n) = \frac{(2n)^q}{q!} \left(1 + \frac{q(q-1)}{4n} + \frac{q(q-1)(3q^2 - 7q - 22)}{96n^2} + O_q(n^{-3}) \right). \quad (6.19)$$

Since D_q is a polynomial, this “asymptotic expansion” terminates after finitely many powers. Formula [Equation \(6.14\)](#) supplies every remaining coefficient algorithmically. The dominant term has a simple combinatorial interpretation: for fixed q , sufficiently many prefix summations make the initial sign $\varepsilon_0 = 1$ dominate the highest-degree part, while later Thue–Morse signs affect only lower powers of n .

6.4 Finite-product and inclusion–exclusion forms

Although $E(z^2)$ is an infinite product, only finitely many factors can affect D_r . If $J = \lfloor \log_2 r \rfloor$ for $r \geq 1$, then

$$D_r(x) = [z^r] \frac{\prod_{j=1}^J (1 - z^{2^j})}{(1 - z)^{2x}}. \quad (6.20)$$

Equivalently, writing $w(S) = \sum_{j \in S} 2^j$,

$$D_r(x) = \sum_{\substack{S \subseteq \{1, \dots, J\} \\ w(S) \leq r}} (-1)^{|S|} \binom{2x + r - w(S) - 1}{r - w(S)}. \quad (6.21)$$

The unique binary representation of each even number identifies [\(6.21\)](#) with [\(1.4\)](#).

6.5 A basis over integer-valued polynomials

The rational monomial coefficients obscure an important integrality fact. Put

$$U(z) = \frac{1}{(1-z)^2} - 1 = 2z + 3z^2 + 4z^3 + \dots$$

Then

$$(1-z)^{-2x} = (1+U(z))^x = \sum_{r \geq 0} \binom{x}{r} U(z)^r.$$

Therefore

$$D_m(x) = \sum_{r=0}^m C_{m,r} \binom{x}{r}, \quad C_{m,r} = [z^m] E(z^2) U(z)^r \in \mathbb{Z}. \quad (6.22)$$

In particular $C_{m,m} = 2^m$, and $D_m(a) \in \mathbb{Z}$ for every integer a , positive or negative. The first few rows $(C_{m,0}, \dots, C_{m,m})$ are

m	binomial-basis coefficients
0	1
1	0, 2
2	-1, 3, 4
3	0, 2, 12, 8
4	-1, 2, 21, 36, 16
5	0, 0, 32, 94, 96, 32
6	1, -1, 41, 195, 328, 240, 64

The low-index coefficients retain the signed Thue–Morse oscillation, while the upper part is governed by the positive Pascal-type factor $(1 - z)^{-2x}$.

Chapter 7

Sheffer structure and efficient recurrences

7.1 A Sheffer family

Define the scaled polynomials

$$\widehat{D}_r(x) = r!D_r(x). \quad (7.1)$$

Then the ordinary generating function in D_r is at the same time the exponential generating function in $\widehat{D}_r$:

$$\sum_{r \geq 0} \frac{\widehat{D}_r(x)}{r!} z^r = \sum_{r \geq 0} D_r(x) z^r = E(z^2) \exp(-2x \log(1 - z)). \quad (7.2)$$

Thus $\widehat{D}_r$ is a Sheffer sequence for the pair

$$A(z) = E(z^2), \quad f(z) = -2 \log(1 - z). \quad (7.3)$$

The terminology and standard operator calculus may be found, for example, in Roman's treatment of the umbral calculus [14]. We derive every identity needed here directly from (7.2).

7.2 Half-step lowering

Theorem 7.1 (Half-step recurrence). *With the convention $D_{-1} = 0$,*

$$\boxed{D_r(x) - D_r(x - \tfrac{1}{2}) = D_{r-1}(x), \quad r \geq 0.} \quad (7.4)$$

Equivalently, for $\widehat{D}_r = r!D_r$,

$$(1 - e^{-\frac{1}{2}\partial_x})\widehat{D}_r(x) = r\widehat{D}_{r-1}(x). \quad (7.5)$$

Proof. Let

$$G_x(z) = \sum_{r \geq 0} D_r(x) z^r = E(z^2)(1 - z)^{-2x}.$$

Then $G_{x-1/2}(z) = (1 - z)G_x(z)$, so $G_x - G_{x-1/2} = zG_x$. Compare coefficients. $\square$

This identity is unusually useful: it says that a half-unit backward difference in the row parameter lowers the diagonal index by one. Iterating,

$$\sum_{j=0}^a (-1)^j \binom{a}{j} D_r\left(x - \frac{j}{2}\right) = D_{r-a}(x), \quad a \geq 0. \quad (7.6)$$

7.3 Differentiation and addition

Theorem 7.2 (Differential recurrence). *For $r \geq 1$,*

$$D'_r(x) = 2 \sum_{j=1}^r \frac{D_{r-j}(x)}{j}. \quad (7.7)$$

Proof. Differentiate $G_x(z)$ with respect to x :

$$\partial_x G_x(z) = -2 \log(1-z) G_x(z) = 2 \left(\sum_{j \geq 1} \frac{z^j}{j} \right) G_x(z).$$

Compare coefficients. □

Theorem 7.3 (Addition law). *For arbitrary x, y ,*

$$D_r(x+y) = \sum_{j=0}^r D_{r-j}(x) \binom{2y+j-1}{j}. \quad (7.8)$$

Proof. Factor $G_{x+y}(z) = G_x(z)(1-z)^{-2y}$ and take the coefficient of z^r . □

The half-step identity is the specialization $y = -1/2$ after moving terms. The addition law also makes clear that translation in x is triangular in the diagonal index.

7.4 The ruler-function recurrence

The most efficient recurrence for generating many diagonal polynomials comes from the logarithm of G_x . Recall that $\nu_2(m)$ is the exponent of 2 in m .

Lemma 7.4 (Ruler logarithm). *One has*

$$\log G_x(z) = \sum_{m \geq 1} \frac{\lambda_m(x)}{m} z^m, \quad \lambda_m(x) = 2x + 2 - 2^{\nu_2(m)+1}. \quad (7.9)$$

Proof. The denominator contributes $-2x \log(1-z) = 2x \sum_{m \geq 1} z^m/m$. On the other hand,

$$\log E(z^2) = \sum_{j \geq 1} \log(1 - z^{2^j}) = - \sum_{j \geq 1} \sum_{a \geq 1} \frac{z^{a2^j}}{a}.$$

For fixed m , the divisors $2^j \mid m$ with $j \geq 1$ contribute

$$-\frac{1}{m} \sum_{j=1}^{\nu_2(m)} 2^j = -\frac{2^{\nu_2(m)+1} - 2}{m}.$$

Adding the two parts gives (7.9). □

Theorem 7.5 (Ruler recurrence). *The diagonal polynomials satisfy*

$$D_0(x) = 1, \quad r D_r(x) = \sum_{m=1}^r (2x + 2 - 2^{\nu_2(m)+1}) D_{r-m}(x). \quad (7.10)$$

Proof. Differentiate $G_x(z) = \exp(\sum c_m z^m / m)$ with respect to z :

$$zG'_x(z) = G_x(z) \sum_{m \geq 1} \lambda_m(x) z^m.$$

Coefficient comparison gives the recurrence. □

The values $2^{\nu_2(m)}$ form the classical ruler sequence. Hence all digital dependence in this polynomial recurrence enters through a one-line valuation statistic rather than explicit binary digit sums.

7.5 Bell-polynomial form

Let $\mathcal{B}_r(y_1, \dots, y_r)$ be the complete exponential Bell polynomial, defined by

$$\exp\left(\sum_{m \geq 1} y_m \frac{z^m}{m!}\right) = \sum_{r \geq 0} \mathcal{B}_r(y_1, \dots, y_r) \frac{z^r}{r!}. \quad (7.11)$$

Then (7.9) gives an immediate closed form.

Corollary 7.6 (Bell-polynomial representation).

$$D_r(x) = \frac{1}{r!} \mathcal{B}_r(0! \lambda_1(x), 1! c_2(x), \dots, (r-1)! \lambda_r(x)), \quad (7.12)$$

where $\lambda_m(x) = 2x + 2 - 2^{\nu_2(m)+1}$.

Bell polynomials provide determinant, partition, and recurrence formulas through standard identities [12]. In computation, (7.10) is usually simpler, while (7.12) provides the cleanest conceptual packaging.

The Bell form (7.12) has an explicit partition expansion and a determinant form; both are standard consequences of the logarithmic data (7.9), with $\lambda_r = \lambda_r(x)$ the ruler cumulants.

If a partition $\mu \vdash m$ has a_r parts equal to r and

$$z_\mu = \prod_{r \geq 1} r^{a_r} a_r!,$$

then

$$D_m(x) = \sum_{\mu \vdash m} \frac{1}{z_\mu} \prod_{r \geq 1} \lambda_r(x)^{a_r}. \quad (7.13)$$

Thus every coefficient is a finite weighted sum indexed by integer partitions, with the weight of a part determined only by its 2-adic valuation.

A determinant form follows from the standard Newton determinant:

$$m! D_m(x) = \det \begin{pmatrix} \lambda_1 & -1 & 0 & \cdots & 0 \\ \lambda_2 & \lambda_1 & -2 & \ddots & \vdots \\ \lambda_3 & \lambda_2 & \lambda_1 & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & -(m-1) \\ \lambda_m & \lambda_{m-1} & \lambda_{m-2} & \cdots & \lambda_1 \end{pmatrix}, \quad (7.14)$$

where $\lambda_r = \lambda_r(x)$. Formulas (7.12)–(7.14) are equivalent descriptions of the same logarithmic data; they are useful in different symbolic systems.

7.6 Translation formulas on the half-grid

The iterated half-step (7.6) is one of three translation formulas that follow from multiplying the generating series by powers of $1 - z$.

Corollary 7.7 (Half-grid translation formulas). *For every integer $a \geq 0$,*

$$D_m\left(x - \frac{a}{2}\right) = \sum_{j=0}^{\min(a,m)} (-1)^j \binom{a}{j} D_{m-j}(x), \quad (7.15)$$

$$D_m\left(x + \frac{1}{2}\right) = \sum_{j=0}^m D_j(x), \quad (7.16)$$

$$D_m(x + 1) = \sum_{j=0}^m (m - j + 1) D_j(x). \quad (7.17)$$

Proof. The first identity follows by iterating multiplication of the generating series by $1 - z$. A positive half-step multiplies it by $(1 - z)^{-1}$, and a unit step by $(1 - z)^{-2}$, giving the other two coefficient convolutions. $\square$

Finally, applying the local table equation (3.8) along a subdiagonal gives the integer-step relation

$$D_m(n) - D_m(n - 1) = 2D_{m-1}(n) - D_{m-2}(n), \quad (7.18)$$

with $D_{-1} = D_{-2} = 0$. This is also obtained by applying the half-step law twice. Finally, applying the Mahler relation one level deeper gives a binary self-similarity for the diagonal generating function itself.

Proposition 7.8 (Diagonal Mahler equation). *If $G_x(z) = \sum_{q \geq 0} D_q(x) z^q$, then*

$$\boxed{G_x(z) = (1 - z)(1 + z)^{2x+1} G_x(z^2)}. \quad (7.19)$$

Equivalently, with

$$g_m(x) = \binom{2x+1}{m} - \binom{2x+1}{m-1} \quad \left(\binom{2x+1}{-1} = 0 \right),$$

one has

$$D_q(x) = \sum_{\substack{0 \leq m \leq q \\ q-m \text{ even}}} g_m(x) D_{(q-m)/2}(x). \quad (7.20)$$

Proof. Use $E(z^2) = (1 - z^2)E(z^4)$ and write

$$G_x(z^2) = \frac{E(z^4)}{(1 - z^2)^{2x}}.$$

A direct simplification gives Equation (7.19). The coefficient of z^m in $(1 - z)(1 + z)^{2x+1}$ is $g_m(x)$; coefficient extraction then gives Equation (7.20). $\square$

Chapter 8

Exact denominators, primitive normalization, and rational zeros

The polynomial formula has surprisingly rigid arithmetic. Although the coefficients of D_m are generally rational, one power of two accounts for the entire coefficient content after multiplication by $m!$.

8.1 A primitive integer normalization

Put

$$c_m = \left\lceil \frac{m}{2} \right\rceil, \quad \tilde{D}_m(x) = \frac{m!}{2^{c_m}} D_m(x). \quad (8.1)$$

Theorem 8.1 (Primitive normalization). *For every $m \geq 0$,*

$$\tilde{D}_m(x) \in \mathbb{Z}[x] \quad \text{and} \quad \text{content}(\tilde{D}_m) = 1. \quad (8.2)$$

Its leading coefficient is

$$[x^m] \tilde{D}_m(x) = 2^{m-c_m} = 2^{\lfloor m/2 \rfloor}. \quad (8.3)$$

Proof. Consider the q th summand of $m!D_m$ in (1.4), and put $p = m - 2q$. Pairing consecutive factors gives

$$(2x)^{\bar{p}} = \prod_{j=0}^{p-1} (2x + j) \in 2^{\lceil p/2 \rceil} \mathbb{Z}[x],$$

because each pair $(2x + 2j)(2x + 2j + 1)$ contributes a factor 2, and an unpaired final even factor contributes one more. The integer $m!/p! = (p+1)(p+2) \cdots m$ is a product of $2q$ consecutive integers and is divisible by at least 2^q . Therefore the whole summand is coefficientwise divisible by

$$2^{\lceil p/2 \rceil + q} = 2^{\lceil (m-2q)/2 \rceil + q} = 2^{c_m}.$$

This proves $\tilde{D}_m \in \mathbb{Z}[x]$.

It remains to prove primitivity. The coefficient of x^{c_m} in the $q = 0$ summand of $\tilde{D}_m$ is the unsigned Stirling number $\begin{bmatrix} m \\ c_m \end{bmatrix}$. This number is odd: modulo 2,

$$x^{\bar{m}} = \prod_{j=0}^{m-1} (x + j) \equiv x^{c_m} (x + 1)^{\lfloor m/2 \rfloor},$$

so the coefficient of x^{c_m} is 1. For $q \geq 1$, the coefficient of x^{c_m} in $(2x)^{\bar{m-2q}}/2^{c_m-q}$ carries an additional factor 2^q whenever that degree exists, and is therefore even. Hence $[x^{c_m}] \tilde{D}_m$ is odd.

The leading coefficient follows from (6.10) and is a pure power of two. Any common divisor of all coefficients must therefore be a power of two, but it must also divide the odd coefficient $[x^{c_m}] \tilde{D}_m$. The content is consequently one. $\square$

8.2 The exact common denominator

Let $\text{den}(P)$ denote the least positive integer d such that $dP \in \mathbb{Z}[x]$.

Corollary 8.2 (Exact denominator). *For every $m \geq 0$,*

$$\boxed{\text{den}(D_m) = \frac{m!}{2^{\min(\nu_2(m!), \lceil m/2 \rceil)}}} \quad (8.4)$$

Using Legendre's identity $\nu_2(m!) = m - s_2(m)$, this is completely explicit in the binary digits of m . For $m \geq 4$ it simplifies to

$$\text{den}(D_m) = \frac{m!}{2^{\lceil m/2 \rceil}}. \quad (8.5)$$

Proof. By Theorem 8.1, $D_m = (2^{c_m}/m!) \tilde{D}_m$ with $\tilde{D}_m$ primitive. Reduce the scalar $2^{c_m}/m!$ to lowest terms. Because $\tilde{D}_m$ has content one, no remaining prime in the denominator can cancel from every coefficient. This gives (8.4). The inequality $\nu_2(m!) \geq \lceil m/2 \rceil$ for $m \geq 4$ follows directly by counting even factors, with the small boundary cases checked separately. $\square$

The first values are

m	0	1	2	3	4	5	6	7
$\text{den}(D_m)$	1	1	1	3	6	15	90	315

m	8	9	10	11	12	13	14	15
$\text{den}(D_m)$	2520	11340	113400	623700	7484400	48648600	681080400	5108103000

This theorem explains why the seemingly ad hoc denominators in the displayed Wolfram Language rules follow a uniform law.

8.3 Every rational zero lies on the half-integer lattice

Theorem 8.3 (Rational-root restriction). *If $\alpha \in \mathbb{Q}$ and $D_m(\alpha) = 0$, then*

$$\alpha \in \frac{1}{2}\mathbb{Z}. \quad (8.6)$$

Proof. The zero set of D_m is the zero set of the primitive integer polynomial $\tilde{D}_m$. By the rational-root theorem and (8.3), the reduced denominator of a nonzero rational root is a power of two. Write such a root as $p/2^e$ with p odd.

Let $\tilde{D}_m(x) = \sum_{r=0}^m a_r x^r$. For $r \geq c_m$, every coefficient of x^r in $m!D_m$ is divisible by 2^r , because the coefficient of x^r in every rising factorial $(2x)^{\overline{p}}$ is divisible by 2^r . Hence

$$2^{r-c_m} \mid a_r \quad (r \geq c_m), \quad (8.7)$$

and equality holds in valuation for the leading coefficient $a_m = 2^{m-c_m}$.

Assume $e \geq 2$ and multiply $\tilde{D}_m(p/2^e) = 0$ by 2^{em} . The leading term has 2-adic valuation $m - c_m$. For $c_m \leq r < m$, the r th term has valuation at least

$$(r - c_m) + e(m - r) = m - c_m + (e - 1)(m - r) > m - c_m.$$

For $r < c_m$, its valuation is at least $e(m - r) \geq 2(m - c_m + 1) > m - c_m$. Thus the leading term is the unique term of least 2-adic valuation, so the sum cannot vanish. This contradiction shows $e \leq 1$. $\square$

The theorem reduces the classification of rational linear factors to exact values on the half-integer lattice. The nonnegative half is completely classified next; the negative half has an exact finite test.

Chapter 9

Exact half-integer roots and factor patterns

The striking linear factors in the first displayed polynomials are not accidental. They are the visible part of an exact modular zero theorem, whose source is a transposition: at a nonnegative half-integer the diagonal generating function becomes an ordinary prefix generating function of a different order.

9.1 Diagonal-prefix transposition

At a nonnegative half-integer, the diagonal generating function becomes an ordinary prefix generating function of a different order. This is the key to the root classification.

Theorem 9.1 (Half-grid transposition). *For every $q, m \in \mathbb{N}$,*

$$\boxed{D_q(m/2) = T_q^{(m+1)}}. \quad (9.1)$$

Thus the polynomial D_q , sampled at $0, \frac{1}{2}, 1, \frac{3}{2}, \dots$, reads the fixed column q vertically through all prefix orders $1, 2, 3, 4, \dots$

Proof. By Equation (7.2) and Equation (2.3),

$$\sum_{q \geq 0} D_q(m/2) z^q = \frac{E(z^2)}{(1-z)^m} = \frac{E(z)}{(1-z)^{m+1}} = \sum_{q \geq 0} T_q^{(m+1)} z^q.$$

Compare coefficients. □

This identity is a useful conceptual inversion. The original table fixes an odd prefix order $2n + 1$ and varies the column. A diagonal polynomial fixes the shifted column q and, at half-integers, varies the prefix order.

9.2 Nonnegative half-integers

The striking linear factors in the first displayed polynomials are not accidental. They are the visible part of an exact modular zero theorem.

Theorem 9.2 (Complete nonnegative half-integer zero criterion). *Let $r, m \in \mathbb{N}_0$, put*

$$h = m + 1, \quad L = 2^h. \quad (9.2)$$

Then

$$\boxed{D_r(m/2) = 0 \iff r \bmod L \in \{L - h, L - h + 1, \dots, L - 1\}}. \quad (9.3)$$

Equivalently,

$$(2x - m) \mid D_r(x) \text{ in } \mathbb{Q}[x] \iff r \bmod 2^{m+1} \geq 2^{m+1} - (m + 1). \quad (9.4)$$

These are exactly all zeros of D_r on $\{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$.

Proof. At $x = m/2$, equation (4.6) gives

$$D_r(m/2) = T_r^{(m+1)} = T_r^{(h)}.$$

Factor the generating function at level h :

$$\begin{aligned} \frac{E(z)}{(1-z)^h} &= \frac{\prod_{j=0}^{h-1} (1 - z^{2^j})}{(1-z)^h} E(z^{2^h}) \\ &= Q_h(z) E(z^L), \end{aligned} \quad (9.5)$$

where

$$Q_h(z) = \prod_{j=0}^{h-1} (1 + z + \dots + z^{2^j-1}). \quad (9.6)$$

This polynomial has degree

$$\sum_{j=0}^{h-1} (2^j - 1) = L - h - 1$$

and every coefficient from degree 0 through $L - h - 1$ is positive. Since $E(z^L) = \sum_{q \geq 0} \varepsilon_q z^{qL}$, the coefficient in each length- L block is a signed copy of the coefficient vector of Q_h , followed by exactly h zeros. This proves (9.3). Distinct values of m give distinct roots, so the corresponding product of linear factors divides D_r . $\square$

The first residue conditions are particularly transparent:

m	root	condition on r	forced factor
0	0	$r \equiv 1 \pmod{2}$	x
1	$1/2$	$r \equiv 2, 3 \pmod{4}$	$2x - 1$
2	1	$r \equiv 5, 6, 7 \pmod{8}$	$x - 1$
3	$3/2$	$r \equiv 12, 13, 14, 15 \pmod{16}$	$2x - 3$
4	2	$r \equiv 27, 28, 29, 30, 31 \pmod{32}$	$x - 2$

For instance, D_7 must contain $x(2x - 1)(x - 1)$, exactly as in (6.8). The additional factor $x + 2$ comes from the negative half-lattice analysis below.

For fixed r , define the systematic nonnegative factor

$$\Phi_r(x) = \prod_{\substack{m \geq 0 \\ r \bmod 2^{m+1} \geq 2^{m+1} - (m+1)}} (2x - m). \quad (9.7)$$

The product is finite because $2^{m+1} - (m+1) > r$ for all sufficiently large m . The theorem says $\Phi_r \mid D_r$ in $\mathbb{Q}[x]$, up to an irrelevant nonzero scalar convention. It also completely identifies the nonnegative half-integer roots, although D_r may have additional negative rational, irrational, or complex roots.

Remark 9.3 (Sign and magnitude of the nonzero values). The proof shows more than the zero criterion. If $r = qL + a$ with $0 \leq a < L - h$, then $D_r(m/2) = \varepsilon_q q_{h,a}$ with $q_{h,a} > 0$ the coefficient of Q_h in (9.6): the sign of a nonzero half-grid value is the Thue–Morse sign of its block index, and its magnitude is a coefficient of the block polynomial studied in Chapter 10. Combined with the rational-root restriction of Theorem 8.3, the criterion is a complete description of the nonnegative rational roots, which the next section spells out.

9.3 Complete classification of nonnegative rational roots

Applying the block zero criterion [Equation \(10.4\)](#) to the right side of [Equation \(9.1\)](#) with order $r = m + 1$ gives [Theorem 9.2](#) a second time; combined with the rational-root restriction it yields the following complete description.

Because every rational root is half-integral by [Theorem 8.3](#), this result is stronger than a grid statement.

Corollary 9.4 (All nonnegative rational roots). *Define*

$$Z_q = \{m \in \mathbb{N} : q \bmod 2^{m+1} \geq 2^{m+1} - (m + 1)\}. \quad (9.8)$$

Then the nonnegative rational roots of D_q are exactly

$$\left\{ \frac{m}{2} : m \in Z_q \right\}. \quad (9.9)$$

In particular,

$$L_q(x) := \prod_{m \in Z_q} (2x - m) \quad (9.10)$$

divides $D_q(x)$ in $\mathbb{Q}[x]$, and L_q contains every linear factor of D_q having a nonnegative rational zero.

The criterion is especially simple in terms of the distance of the low binary suffix from an all-ones word. If

$$b = q \bmod 2^{m+1}, \quad \delta_m(q) = 2^{m+1} - 1 - b,$$

then $m \in Z_q$ exactly when $\delta_m(q) \leq m$.

Proposition 9.5 (Number of nonnegative rational roots). *For $q \geq 1$,*

$$|Z_q| \leq \lceil \log_2(q + 1) \rceil. \quad (9.11)$$

The bound is attained at every $q = 2^a - 1$, where

$$Z_{2^a-1} = \{0, 1, \dots, a - 1\} \quad (9.12)$$

and hence

$$\prod_{m=0}^{a-1} (2x - m) \mid D_{2^a-1}(x). \quad (9.13)$$

More generally, if $q = 2^a - 1 - \delta$ with $0 \leq \delta \leq a - 1$, then

$$\{\delta, \delta + 1, \dots, a - 1\} \subseteq Z_q. \quad (9.14)$$

Proof. Put $a = \lceil \log_2(q + 1) \rceil$, so $q < 2^a$. For $m \geq a$,

$$2^{m+1} - (m + 1) \geq 2^{a+1} - (a + 1) > 2^a - 1 \geq q,$$

so the root criterion cannot hold. This proves [Equation \(9.11\)](#).

If $q = 2^a - 1$ and $m < a$, its residue modulo 2^{m+1} is $2^{m+1} - 1$, which is in the zero window. No $m \geq a$ occurs by the previous paragraph, proving [Equation \(9.12\)](#).

Finally, if $q = 2^a - 1 - \delta$ and $\delta \leq m \leq a - 1$, then

$$q \bmod 2^{m+1} = 2^{m+1} - 1 - \delta \geq 2^{m+1} - (m + 1),$$

which proves [Equation \(9.14\)](#). □

The criterion also detects irrational positive roots by sign bracketing. For example,

$$D_4(1/2) = -1, \quad D_4(1) = 1,$$

while $Z_4 = \emptyset$. Thus D_4 has a root in $(1/2, 1)$, and [Theorem 8.3](#) proves that this root is irrational. Numerically it is approximately 0.8440034891960957.

9.4 Negative half-integers as finite differences

Extend ε_j by zero for $j < 0$, and let the backward difference be

$$(\nabla f)_r = f_r - f_{r-1}. \quad (9.15)$$

Theorem 9.6 (Negative half-lattice formula). *For every integer $a \geq 1$,*

$$D_r(-a/2) = \sum_{j=0}^{a-1} (-1)^j \binom{a-1}{j} \varepsilon_{r-j} = \nabla^{a-1} \varepsilon_r. \quad (9.16)$$

Proof. Use the alternate generating function

$$\sum_{r \geq 0} D_r(x) z^r = \frac{E(z)}{(1-z)^{2x+1}}.$$

At $x = -a/2$, this becomes $(1-z)^{a-1} E(z)$, whose coefficient is the stated backward difference. $\square$

Several consequences explain further factors in the initial list.

Corollary 9.7 (Two valuation-controlled negative roots). *For $r \geq 1$,*

$$D_r(-1) = 0 \iff \nu_2(r) \text{ is odd.} \quad (9.17)$$

For $r \geq 3$,

$$D_r(-2) = 0 \iff \nu_2(r-1) \text{ is odd.} \quad (9.18)$$

Thus $x+1$ divides D_r under the first condition, and $x+2$ divides it under the second.

Proof. For $x = -1$, formula (9.16) is $\varepsilon_r - \varepsilon_{r-1}$. Incrementing $r-1$ to r changes the binary digit sum by $1 - \nu_2(r)$; hence the two signs agree exactly when $\nu_2(r)$ is odd.

For $x = -2$, the third difference is

$$\varepsilon_r - 3\varepsilon_{r-1} + 3\varepsilon_{r-2} - \varepsilon_{r-3}.$$

A four-sign check shows this vanishes exactly when $\varepsilon_r = \varepsilon_{r-3}$ and $\varepsilon_{r-1} = \varepsilon_{r-2}$. An even r cannot satisfy the second equality. If $r = 2q + 1$, both equalities reduce to $\varepsilon_q = -\varepsilon_{q-1}$, which holds exactly when $\nu_2(q)$ is even. Since $\nu_2(r-1) = 1 + \nu_2(q)$, this is equivalent to (9.18). $\square$

For any fixed a , equation (9.16) reduces the factor question at $x = -a/2$ to a finite local pattern in the Thue–Morse word. This is an effective exact test even when a simple valuation criterion is not available.

9.5 Strict negative half-integers and a conjecture

In the parity-split form used by the second report, the negative half-lattice formula reads, for $b, m \in \mathbb{N}$,

$$D_m\left(-\frac{b}{2}\right) = \sum_{\substack{0 \leq j \leq \min(b,m) \\ j \equiv m \pmod{2}}} (-1)^j \binom{b}{j} \varepsilon_{(m-j)/2}, \quad (9.19)$$

obtained by expanding $(1-z)^b E(z^2)$ instead of $(1-z)^{b-1} E(z)$; the two expansions agree by the Mahler identity. Some low negative arguments simplify sharply: $D_m(-1/2) = \varepsilon_m$, $D_{2q}(-1) = \varepsilon_q + \varepsilon_{q-1}$, $D_{2q+1}(-1) = -2\varepsilon_q$, $D_{2q}(-2) = \varepsilon_q + 6\varepsilon_{q-1} + \varepsilon_{q-2}$ and $D_{2q+1}(-2) = -4(\varepsilon_q + \varepsilon_{q-1})$, with $\varepsilon_j = 0$ for $j < 0$; the criteria of Theorem 9.7 are these values read through $\varepsilon_q = -\varepsilon_{q-1} \iff \nu_2(q)$ even.

The first two strict negative half-integers are also never roots: $D_m(-1/2) = \varepsilon_m \neq 0$, and

$$D_m(-3/2) = \varepsilon_m - 2\varepsilon_{m-1} + \varepsilon_{m-2} \neq 0.$$

For $m \geq 2$, the second assertion would require three consecutive Thue–Morse signs to be equal, while every three consecutive positions contain an aligned pair $(2q, 2q + 1)$ of opposite signs. The cases $m = 0, 1$ are immediate from the same formula with the zero extension just stated.

The experiments included with this volume suggest a stronger statement.

Conjecture 9.8 (No strict negative half-integer roots). *For all $m, r \in \mathbb{N}$,*

$$D_m\left(-r - \frac{1}{2}\right) \neq 0. \quad (9.20)$$

Equivalently, every negative rational root of D_m is an integer.

The accompanying exact Python program found no counterexample for every odd $1 \leq b \leq 63$ and every $0 \leq m \leq 10000$ in $D_m(-b/2)$. This finite computation is evidence only; the conjecture is not used elsewhere. Formula (9.19) converts the problem into a nonvanishing statement for parity-restricted binomial sums of consecutive Thue–Morse signs.

9.6 A provable family of negative integer roots

The nonnegative rational roots are completely classified. Negative roots have a richer structure. The following theorem gives an infinite exact family; it is deliberately stated as a sufficient condition rather than a complete classification.

Theorem 9.9 (Trailing-ones negative roots). *Suppose $q \equiv -1 \pmod{2^a}$. Let ℓ be odd and satisfy $1 \leq \ell \leq a - 1$. Then*

$$D_q(-2^\ell) = 0. \quad (9.21)$$

In particular, for $q = 2^a - 1$,

$$\prod_{\substack{1 \leq \ell \leq a-1 \\ \ell \text{ odd}}} (x + 2^\ell) \text{ divides } D_{2^a-1}(x). \quad (9.22)$$

Combining with Equation (9.13),

$$\left(\prod_{m=0}^{a-1} (2x - m) \right) \left(\prod_{\substack{1 \leq \ell \leq a-1 \\ \ell \text{ odd}}} (x + 2^\ell) \right) \mid D_{2^a-1}(x). \quad (9.23)$$

Proof. Put $h = 2^\ell$. At $x = -h$, the diagonal generating function and Equation (2.3) give

$$\sum_{q \geq 0} D_q(-h)z^q = E(z^2)(1 - z)^{2h} = E(z)(1 - z)^{2h-1}.$$

Therefore

$$D_q(-h) = \sum_{j=0}^{2h-1} (-1)^j \binom{2h-1}{j} \varepsilon_{q-j}, \quad (9.24)$$

where terms with $j > q$ are omitted. The hypothesis $\ell \leq a - 1$ gives $2h - 1 < 2^a$. Write $q = Q2^a + (2^a - 1)$. Subtracting any $0 \leq j < 2^a$ affects only the low a bits, so binary complementation gives

$$\varepsilon_{q-j} = \varepsilon_Q \varepsilon_{2^a-1-j} = \varepsilon_Q (-1)^a \varepsilon_j.$$

The fixed nonzero factor $\varepsilon_Q(-1)^a$ may be ignored. Also,

$$(-1)^j \varepsilon_j = \varepsilon_{\lfloor j/2 \rfloor},$$

by checking the even and odd cases with [Equation \(2.1\)](#). Pairing $j = 2r$ and $j = 2r+1$ in [Equation \(9.24\)](#) reduces the remaining sum to

$$\sum_{r=0}^{h-1} \varepsilon_r \binom{2h}{2r+1}. \quad (9.25)$$

Since $h = 2^\ell$, the map $r \mapsto h-1-r$ complements all ℓ binary digits. Thus $\varepsilon_{h-1-r} = (-1)^\ell \varepsilon_r = -\varepsilon_r$, while

$$\binom{2h}{2(h-1-r)+1} = \binom{2h}{2r+1}.$$

The terms of [Equation \(9.25\)](#) cancel in pairs. □

The theorem explains, for example, the factors $x+2$ in $D_3, D_7, D_{15}, \dots$, and the additional factor $x+8$ in $D_{15}, D_{31}, \dots$. Computation reveals further negative integer factors not covered by [Theorem 9.9](#); their complete binary classification is a natural open problem discussed in [Chapter 15](#).

Chapter 10

Finite dyadic blocks for arbitrary summation order

This section proves the global periodic shape from which both the row acceleration rules and the half-integer root theorem arise.

10.1 The block polynomial

Fix an integer $h \geq 1$ and set

$$L = 2^h, \quad P_h(z) = \prod_{j=0}^{h-1} (1 - z^{2^j}), \quad (10.1)$$

so that $E(z) = P_h(z)E(z^L)$. Because

$$1 - z^{2^j} = (1 - z)(1 + z + \cdots + z^{2^j-1}),$$

define

$$Q_h(z) = \frac{P_h(z)}{(1 - z)^h} = \prod_{j=0}^{h-1} (1 + z + \cdots + z^{2^j-1}) = \sum_{a=0}^{L-h-1} q_{h,a} z^a. \quad (10.2)$$

The factor for $j = 0$ is simply 1. All coefficients $q_{h,a}$ in the displayed range are positive integers.

Theorem 10.1 (Complete finite-block theorem). *Let $T_r^{(h)}$ be the h -fold inclusive prefix of the signed Thue–Morse sequence. Extend $q_{h,a} = 0$ for $L - h \leq a \leq L - 1$. Then:*

1. **Signed blocks.** For $q \geq 0$ and $0 \leq a < L$,

$$\boxed{T_{qL+a}^{(h)} = \varepsilon_q q_{h,a}.} \quad (10.3)$$

Hence $|T_r^{(h)}|$ is periodic with period L .

2. **Support and terminal zeros.**

$$q_{h,a} > 0 \quad (0 \leq a \leq L - h - 1), \quad q_{h,a} = 0 \quad (L - h \leq a < L). \quad (10.4)$$

Thus every block has exactly h zero positions.

3. **Reflection.**

$$q_{h,a} = q_{h,L-h-1-a}, \quad 0 \leq a \leq L - h - 1. \quad (10.5)$$

4. **Maximum and plateau.** The maximum is

$$M_h = 2^{\binom{h-1}{2}}, \quad (10.6)$$

and it occurs exactly at

$$a = L/2 - h, L/2 - h + 1, \dots, L/2 - 1, \quad (10.7)$$

a plateau of length h .

5. **Strict ramps.** Before the plateau the positive values are strictly increasing; after it they are their reflected strictly decreasing copy.

6. **Mass.**

$$\sum_{a=0}^{L-1} q_{h,a} = 2^{\binom{h}{2}}. \quad (10.8)$$

7. **Multiplicity and distinct values.** Zero and M_h each occur exactly h times. Every other value occurs exactly twice. The number of distinct values is

$$\boxed{2^{h-1} - h + 2}. \quad (10.9)$$

8. **Complement pairing.** Put $d = L/2 - h$. For $0 \leq a < d$,

$$\boxed{q_{h,a} + q_{h,d-1-a} = M_h}. \quad (10.10)$$

Thus every value present in the block has its complement relative to M_h present as well.

Proof. From (2.8) and (10.2),

$$\frac{E(z)}{(1-z)^h} = Q_h(z)E(z^L). \quad (10.11)$$

Since $\deg Q_h = L - h - 1 < L$ and $E(z^L) = \sum_{q \geq 0} \varepsilon_q z^{qL}$, coefficient extraction immediately gives the signed-block law. The degree calculation gives the terminal zero run. The product in (10.2) has positive coefficients at every degree from zero through its degree, because it counts bounded compositions with interval-valued parts. This proves support.

Each geometric factor is reciprocal, hence Q_h is reciprocal of degree $L - h - 1$, proving reflection. To analyze the shape, split off the largest factor:

$$Q_h(z) = C_h(z)(1 + z + \dots + z^{L/2-1}), \quad C_h(z) = \prod_{j=0}^{h-2} (1 + z + \dots + z^{2^j-1}). \quad (10.12)$$

The polynomial C_h has positive coefficients on the full interval

$$0 \leq t \leq \deg C_h = L/2 - h \quad (10.13)$$

and total coefficient mass

$$C_h(1) = \prod_{j=0}^{h-2} 2^j = 2^{\binom{h-1}{2}} = M_h. \quad (10.14)$$

For $0 \leq a < L/2$, the coefficient $q_{h,a}$ is the cumulative sum of the coefficients of C_h through degree a . It therefore increases strictly until all coefficients have entered the window, at $a = L/2 - h$, and then remains equal to M_h through $a = L/2 - 1$. Reciprocity supplies the decreasing half and proves the plateau assertion.

Evaluating $Q_h(1)$ gives $\prod_{j=0}^{h-1} 2^j = 2^{\binom{h}{2}}$, proving the mass. The strict ramps and reflection show that every nonmaximum positive value occurs exactly twice. There are h terminal zeros and h maximum positions, so the left strict ramp has $L/2 - h$ entries. Counting one value per left-ramp position together with zero and the maximum gives $L/2 - h + 2$, which is (10.9).

Finally, C_h is reciprocal of degree $d = L/2 - h$. A prefix of its coefficient sequence through a and the reflected prefix through $d - 1 - a$ partition the total mass M_h , giving (10.10). $\square$

The period, block sum, maximum, number of zeros, number of maxima, distinct value count, double multiplicity, and complement property in this theorem are exactly the phenomena recorded experimentally in the 2018 question [7]. The proof shows that they are all shadows of the same finite factorization (10.11).

10.2 The block as a probability mass function

After division by its mass, the block is the distribution of a sum of independent discrete uniforms, which gives its moments in closed form.

Proposition 10.2 (Moments of the block). *For $h \geq 1$, the normalized coefficients $q_{h,a}/2^{\binom{h}{2}}$, $0 \leq a \leq L - h - 1$, form the probability mass function of*

$$U_1 + \cdots + U_{h-1}, \quad U_j \sim \text{Unif}\{0, 1, \dots, 2^j - 1\},$$

with independent summands. Therefore

$$\mathbb{E}\left[\sum_j U_j\right] = \frac{L - h - 1}{2}, \quad (10.15)$$

$$\text{Var}\left[\sum_j U_j\right] = \frac{4^h - 3h - 1}{36}. \quad (10.16)$$

Proof. Divide every factor of (10.2) by its value 2^j at $z = 1$; the j th factor becomes the probability generating function of U_j , and the product is that of the independent sum. The mean of U_j is $(2^j - 1)/2$ and its variance $(4^j - 1)/12$; summing over $1 \leq j \leq h - 1$ with the geometric-series formulas gives the two displays. $\square$

The mean is the center of the palindromic block, as it must be. These moments reappear in Chapter 12: after the dyadic scaling the variance tends to the variance $1/36$ of the uniform sum underlying the up-function.

Chapter 11

Specialization to the rows of the table

11.1 One positive block and signed repetition

For row n , set

$$h = 2n + 1, \quad L_n = 2^{2n+1}. \quad (11.1)$$

The shifted row generating function (3.3) and the block factorization give

$$A_n(z) = B_n(z)E(z^{L_n}), \quad (11.2)$$

where

$$B_n(z) = z^{n+1}Q_{2n+1}(z) = z^{n+1} \prod_{j=1}^{2n} (1 + z + \cdots + z^{2^j-1}). \quad (11.3)$$

Write

$$B_n(z) = \sum_{a=0}^{L_n-1} b_{n,a} z^a, \quad (11.4)$$

with trailing zero coefficients included to make a complete block.

Theorem 11.1 (Exact row block law). *For every $n, q \geq 0$ and $0 \leq a < L_n$,*

$$s_{n,qL_n+a} = \varepsilon_q b_{n,a}. \quad (11.5)$$

In particular, $|s_{n,k}|$ is periodic with period L_n , while the signs of consecutive blocks follow the signed Thue–Morse sequence.

Proof. This is coefficient extraction from (11.2) and $E(z^{L_n}) = \sum_{q \geq 0} \varepsilon_q z^{qL_n}$. □

This proves the first large-index acceleration branch in the supplied code:

$$s_{n,k} = \varepsilon_{\lfloor k/L_n \rfloor} s_{n,k \bmod L_n}.$$

11.2 Complete geometry of the first block

Let

$$M_n = 2^{\binom{2n}{2}} = 2^{n(2n-1)}. \quad (11.6)$$

Shifting the arbitrary-order theorem yields the following row-specific form.

Theorem 11.2 (Geometry of row n). *For $0 \leq a < L_n$, the first block $b_{n,a}$ satisfies:*

1. **Support.**

$$b_{n,a} > 0 \iff n+1 \leq a \leq L_n - n - 1. \quad (11.7)$$

There are exactly $2n+1$ zero positions.

2. **Circular mirror.** With $b_{n,L_n} = b_{n,0} = 0$,

$$b_{n,a} = b_{n,L_n-a}, \quad 0 \leq a \leq L_n. \quad (11.8)$$

3. **Half-block complement.** For $0 \leq a \leq L_n/2$,

$$b_{n,a} + b_{n,L_n/2-a} = M_n. \quad (11.9)$$

In particular,

$$b_{n,L_n/4} = M_n/2 \quad (n \geq 1). \quad (11.10)$$

4. **Central plateau.**

$$b_{n,a} = M_n \iff L_n/2 - n \leq a \leq L_n/2 + n. \quad (11.11)$$

There are exactly $2n+1$ maximum positions.

5. **Strict ramps.** After the initial zeros, the block increases strictly to the plateau; after the plateau it decreases strictly to the final zeros.6. **Mass.**

$$\sum_{a=0}^{L_n-1} b_{n,a} = 2^{n(2n+1)}. \quad (11.12)$$

7. **Multiplicity and distinct values.** Every value other than zero and M_n occurs exactly twice. The number of distinct values is

$$2^{2n} - 2n + 1. \quad (11.13)$$

Proof. The relation between the row profile and the general block is

$$b_{n,a} = q_{2n+1,a-n-1}, \quad (11.14)$$

where $q_{h,t} = 0$ outside its natural support. Translating the support, reflection, plateau, mass, and multiplicity statements from the preceding theorem gives every assertion except the convenient form (11.9). The general complement relation translates as

$$(a - n - 1) + (L_n/2 - a - n - 1) = L_n/2 - (2n + 1) - 1,$$

which is precisely the pair of reciprocal cumulative positions in the proof of (10.10). The endpoint zero/maximum cases extend the identity to the full interval $0 \leq a \leq L_n/2$. $\square$

The mirror identity (11.8) proves the second dyadic branch in the supplied code. The complement identity (11.9) proves the third branch and strengthens its stated range from the second quarter to the entire first half. The strict lower bound in the program is necessary only to avoid a self-reference at $a = L_n/4$; equation (11.10) supplies that missing value directly.

11.3 Bounded-composition interpretation

The coefficient $b_{n,a}$ counts bounded compositions:

$$b_{n,a} = \# \left\{ (y_1, \dots, y_{2n}) : 0 \leq y_j < 2^j, \sum_{j=1}^{2n} y_j = a - n - 1 \right\}. \quad (11.15)$$

Consequently, whenever the diagonal position lies inside the first positive block,

$$D_r(n) = \# \left\{ (y_1, \dots, y_{2n}) : 0 \leq y_j < 2^j, \sum_{j=1}^{2n} y_j = r \right\}. \quad (11.16)$$

For every fixed r , this applies for all sufficiently large n . Thus the diagonal polynomial eventually counts a family of bounded compositions, while its evaluations at small n may retain signs inherited from later Thue–Morse blocks.

11.4 Normalized profiles and the Fabius connection

Figure 11.1 displays the normalized block

$$\frac{b_{n,a}}{M_n} \quad \text{against} \quad \frac{a}{L_n}, \quad (11.17)$$

for the first few rows. The repository develops the normalization and polygonal-interpolant limit that connects these finite coefficient grids to the Fabius function, while the same finite products become sinc products on the Fourier side and lead to the Rvachev up-function [1, 11]. The present volume needs only the exact finite algebra; the figure is included to show the geometric meaning of the row identities.

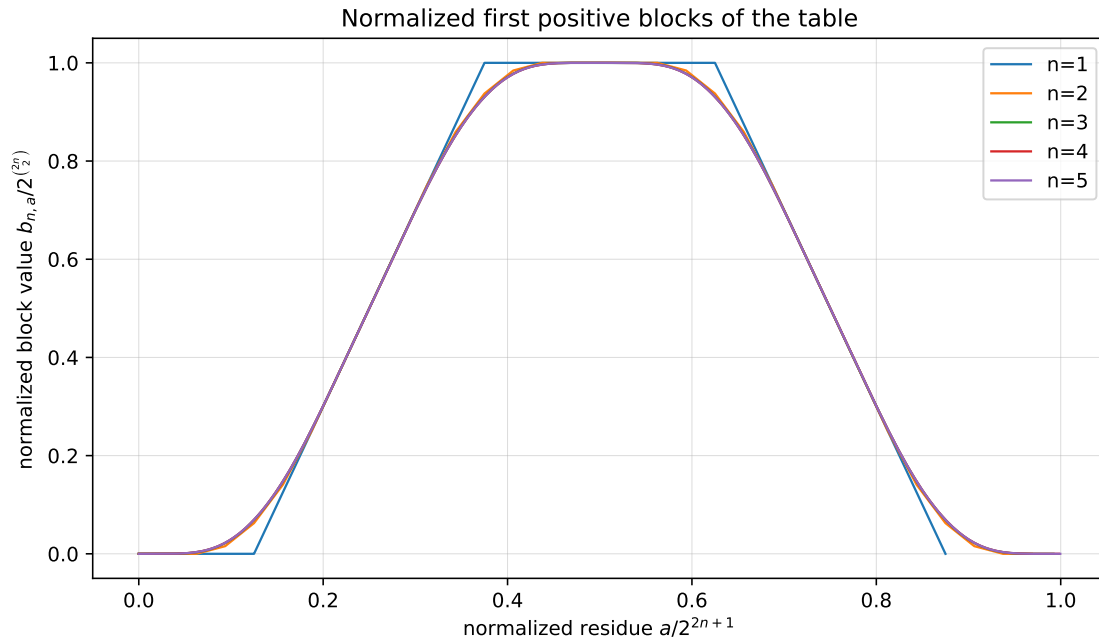


Figure 11.1: Normalized first positive blocks for $n = 1, \dots, 5$, generated by `experiments.py`. The central plateau has exact width $(2n + 1)/2^{2n+1}$, and the mirror/complement laws hold before taking a limit.

11.5 The first nontrivial blocks and a table of the diagonal array

In the second report's notation, $H_a(z) = Q_{a+1}(z)$ is the block polynomial of order $a + 1$, $B = 2^{2n+1} = L_n$ the block length of row n , and the diagonal-offset coefficients are $D_r(n) = \varepsilon_q q_{2n+1,a}$ for $r = qL_n + a$.

Example 11.3 (The first nontrivial finite blocks). For $n = 1$, $B = 8$ and

$$H_2(z) = (1 + z)(1 + z + z^2 + z^3) = 1 + 2z + 2z^2 + 2z^3 + z^4.$$

Thus the diagonal-offset block is

$$1, 2, 2, 2, 1, 0, 0, 0,$$

followed by its negative, then by blocks with signs $-1, +1, \dots$ according to ε_q . For $n = 2$, $B = 32$ and the plateau height is $2^{\binom{4}{2}} = 64$, attained at diagonal offsets 11, 12, 13, 14, 15. (The final offsets 27, 28, 29, 30, 31 are instead the zero window.)

A small portion of the diagonal-coordinate table $(D_m(n))_{n,m \geq 0}$ is displayed below. Rows $n = 0$ and $n = 1$ already show the signed block repetition, while larger rows have not yet reached their first plateau within the displayed window.

$n \backslash m$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	1	0	-1	0	-1	0	1	0	-1	0	1	0	1	0	-1	0
1	1	2	2	2	1	0	0	0	-1	-2	-2	-2	-1	0	0	0
2	1	4	9	16	24	32	40	48	55	60	63	64	64	64	64	64
3	1	6	20	50	104	190	316	490	719	1008	1360	1776	2256	2800	3408	4080
4	1	8	35	112	293	664	1351	2528	4424	7328	11592	17632	25928	37024	51528	70112

Chapter 12

Connection with the Fabius and Rvachev approximation layer

The diagonal theory is purely discrete, but it lands exactly on the finite polynomials used in the repository’s Fabius–Rvachev approximants. The Lean formalization proves that the order- k iterated prefix coefficients below the first dyadic cutoff are the coefficients of the approximation polynomial p_{k-1} , and that division by $2^{\binom{k-1}{2}}$ gives the corresponding step-approximant values at cell centers [5].

For the odd order $k = 2n + 1$ considered here, the finite approximation polynomial is precisely

$$p_{2n}(z) = H_{2n}(z) = \prod_{j=1}^{2n} (1 + z + \cdots + z^{2^j-1}). \quad (12.1)$$

Therefore, whenever $0 \leq m < 2^{2n+1}$,

$$D_m(n) = [z^m]p_{2n}(z), \quad (12.2)$$

and the normalized step height is

$$\frac{D_m(n)}{2^{\binom{2n}{2}}}. \quad (12.3)$$

The normalizing denominator is exactly the plateau height C in (11.6). Thus the complement law (11.9) becomes a normalized reflection around $1/2$, and the central plateau (11.11) becomes the level 1.

For a fixed diagonal offset m , combine (6.19) with $C = 2^{n(2n-1)}$ to obtain

$$\frac{D_m(n)}{2^{\binom{2n}{2}}} = \frac{2^m n^m}{m!} 2^{-n(2n-1)} \left(1 + \frac{m(m-1)}{4n} + \mathcal{O}(n^{-2}) \right). \quad (12.4)$$

Hence every fixed number of coefficient cells from the left edge is suppressed by a quadratic exponential in n , up to a polynomial factor. This is an exact quantitative boundary-layer statement for the finite approximants. It is consistent with the infinite-order flatness of the limiting Fabius/Rvachev functions, but it should not be confused with an asymptotic at a fixed spatial argument: the cell represented by a fixed m moves toward the support endpoint as the grid is refined.

The block factorization also clarifies why the early repeated-summation picture sees the Fabius shape at all. After division by $H_{2n}(1)$, the coefficients of H_{2n} are the probability mass function of a sum of independent discrete uniforms on $\{0, \dots, 2^j - 1\}$, while the normalized limit is related to the continuous sum of geometrically scaled uniforms appearing in the probability model of Rvachev’s up-function. The present volume does not reprove that analytic limit; it isolates the exact finite coefficient algebra on which that limit acts.

12.1 Exact row–histogram identification and the table limits

The identification is exact at the level of cell samples, not only in the limit; in the following, the pulse polynomial of order r is Q_r of (10.2), with coefficients $a_{r,b} = q_{r,b}$.

The finite pulse occurring in row n is not merely analogous to the approximants in the Fabius–Rvachev theory: it is exactly the even-indexed subsequence of those approximants. This identifies the correct shift and normalization of the table without guesswork.

Let up denote Rvachev’s even, compactly supported function, normalized by $\text{up}(0) = 1$, and let F denote the bounded Fabius function on $[0, 1]$, so that

$$F(x) = \text{up}(x - 1) \quad (0 \leq x \leq 1). \quad (12.5)$$

These are the classical functions introduced from different viewpoints by Fabius and the Rvachevs; their arithmetic relation with signed Thue–Morse prefix sums is developed in detail in [13, 15, 10, 2].

For the table row n , retain

$$B_n = 2^{2n+1}, \quad H_n = 2^{2n}, \quad M_n = 2^{\binom{2n}{2}}. \quad (12.6)$$

The polynomial Q_{2n+1} of Equation (10.2) is precisely the finite step polynomial usually denoted $\mathcal{P}_{2n}$. Its degree is

$$G_n = B_n - 2n - 2. \quad (12.7)$$

For $0 \leq m \leq G_n$, put $k = n + 1 + m$. The standard centered cell coordinate attached to coefficient m is

$$\xi_{n,m} = \frac{2m - G_n}{B_n} = \frac{2k - B_n}{B_n}. \quad (12.8)$$

Thus the shift by $n + 1$ in Equation (3.4) exactly cancels the centering correction in the continuous approximation.

Proposition 12.1 (Exact row–histogram identification). *At every coefficient center $\xi_{n,m}$, the normalized centered histogram approximant $\mathcal{W}_{2n}$ satisfies*

$$\mathcal{W}_{2n}(\xi_{n,m}) = \frac{a_{2n+1,m}}{M_n} = \frac{s_{n,n+1+m}}{M_n}. \quad (12.9)$$

In particular, the positive part of the first table block, after division by its exact height M_n , is the level- $2n$ Rvachev histogram sampled at its cell centers.

Proof. The coefficient identity $a_{2n+1,m} = T_m^{(2n+1)}$ in the first dyadic block follows from Equation (10.3). The table identity $s_{n,n+1+m} = T_m^{(2n+1)}$ is Equation (3.4). Finally, the histogram normalization at level $N = 2n$ is

$$\frac{2^N}{2^{\binom{N+1}{2}}} = \frac{1}{2^{\binom{N}{2}}} = \frac{1}{M_n},$$

and its center formula reduces to Equation (12.8). $\square$

The relation can be stated without introducing the intervening histogram. It gives a direct approximation algorithm using only the table itself.

Theorem 12.2 (Table limits to up and F). *For $0 \leq x \leq 1$, define*

$$R_n(x) = \frac{s_{n,n+1+\lfloor B_n x \rfloor}}{M_n}, \quad (12.10)$$

$$F_n(x) = \frac{s_{n,n+1+\lfloor H_n x \rfloor}}{M_n}. \quad (12.11)$$

Then, pointwise,

$$R_n(x) \longrightarrow \text{up}(2x - 1), \quad (12.12)$$

$$F_n(x) \longrightarrow F(x). \quad (12.13)$$

No assertion of a uniform convergence rate is implicit in these formulas.

Proof. By Equation (3.4),

$$R_n(x) = \frac{T_{\lfloor 2^{2n+1}x \rfloor}^{(2n+1)}}{2^{\binom{2n}{2}}}.$$

This is exactly the corrected prefix-grid sample from the repository at level $2n$. The proved corrected convergence theorem there gives $R_n(x) \rightarrow \text{up}(2x - 1)$ on $[0, 1]$ [2]. Since $F_n(x) = R_n(x/2)$, relation Equation (12.5) gives

$$F_n(x) \longrightarrow \text{up}(x - 1) = F(x).$$

□

This theorem clarifies the roles of the two limits studied in the volume. A fixed row and varying column exhibit the compact pulse and its signed block repetitions. A fixed diagonal and varying row produce the polynomials $D_q(n)$. The Fabius limit instead lets the shifted column q grow exponentially with the row, at the scale $q \asymp 2^{2n}$. Thus fixed-diagonal asymptotics such as Equation (6.19) and the continuum limit are complementary scaling regimes of the same exact array.

Chapter 13

Algorithms and Wolfram Language implementation

13.1 Replacing the finite list of diagonal rules

The eight hand-written diagonal branches can be replaced by one exact definition. The following Wolfram Language code uses Pochhammer for the rising factorial and memoizes each symbolic polynomial.

Listing 13.1: Exact polynomial for every diagonal.

```
ClearAll[tmSign, diagonalPolynomial, x];

tmSign[q_Integer?NonNegative] := (-1)^ThueMorse[q];

diagonalPolynomial[0] = 1;
diagonalPolynomial[r_Integer?Positive] :=
  diagonalPolynomial[r] =
    Expand@Sum[
      tmSign[q] Pochhammer[2 x, r - 2 q]/Factorial[r - 2 q],
      {q, 0, Quotient[r, 2]}
    ];

ClearAll[sByDiagonal];
sByDiagonal[n_Integer?NonNegative, k_Integer?NonNegative] /; k <= n := 0;
sByDiagonal[n_Integer?NonNegative, k_Integer?NonNegative] :=
  diagonalPolynomial[k - n - 1] /. x -> n;
```

This route is ideal when $k - n$ is small and n is large: its work depends on the diagonal offset, not on the row index. A direct numeric form avoids symbolic expansion:

Listing 13.2: Direct exact evaluation of one diagonal entry.

```
ClearAll[sDiagonalValue];
sDiagonalValue[n_Integer?NonNegative, k_Integer?NonNegative] /; k <= n := 0;
sDiagonalValue[n_Integer?NonNegative, k_Integer?NonNegative] := Module[
  {r = k - n - 1},
  Sum[
    (-1)^ThueMorse[q] Binomial[2 n + r - 2 q - 1, r - 2 q],
    {q, 0, Quotient[r, 2]}
  ]
];
```

13.2 Generating consecutive polynomials by the ruler recurrence

When all diagonals through $r = R$ are wanted, the recurrence (7.10) reuses previous results and exposes the valuation structure.

Listing 13.3: Ruler-function recurrence for consecutive diagonals.

```
ClearAll[diagonalRuler, x];
diagonalRuler[0] = 1;
diagonalRuler[r_Integer?Positive] :=
  diagonalRuler[r] = Expand[
    Sum[
      (2 x + 2 - 2^(IntegerExponent[m, 2] + 1))
        diagonalRuler[r - m],
      {m, 1, r}
    ]/r
  ];
```

The recurrence is triangular and requires $R(R+1)/2$ polynomial multiplications by linear factors. For one isolated diagonal, (1.4) is usually shorter; for a full prefix of the family, the ruler recurrence is usually preferable.

13.3 Precomputing a complete row block

For fixed moderate n and many arbitrary k , precompute the one positive block B_n and use the signed block law.

Listing 13.4: Block-based exact table lookup.

```
ClearAll[z, rowBlockPolynomial, rowBlock, sByBlock];

rowBlockPolynomial[n_Integer?NonNegative] :=
  rowBlockPolynomial[n] = Expand[
    z^(n + 1) Product[
      Sum[z^a, {a, 0, 2^j - 1}],
      {j, 1, 2 n}
    ]
  ];

rowBlock[n_Integer?NonNegative] := rowBlock[n] = Module[
  {len = 2^(2 n + 1)},
  PadRight[CoefficientList[rowBlockPolynomial[n], z], len]
];

sByBlock[n_Integer?NonNegative, k_Integer?NonNegative] := Module[
  {len = 2^(2 n + 1), q, a},
  q = Quotient[k, len];
  a = Mod[k, len];
  (-1)^ThueMorse[q] rowBlock[n][[a + 1]]
];
```

The polynomial expansion shown here is concise and suitable for presentation. For larger n , the companion Python code computes the same coefficient list by repeated convolution with all-one windows, using a sliding sum so each factor is processed in linear time in the current output length.

13.4 Useful additional acceleration rules

The structural theorem supplies exact branches that are absent from the original listing. Inside the first block and for $n \geq 1$,

$$s_{n, L_n/4} = M_n/2, \quad (13.1)$$

$$s_{n,a} = M_n \quad \text{for } L_n/2 - n \leq a \leq L_n/2 + n, \quad (13.2)$$

$$s_{n,a} = 0 \quad \text{for } 0 \leq a \leq n \text{ or } L_n - n \leq a < L_n. \quad (13.3)$$

Together with the stronger full-half complement (11.9), these can reduce recursion depth substantially. Care is required to order conditional definitions so that fixed points such as $a = L_n/4$ are handled before a symmetry branch referring to themselves.

The standalone file `thue_morse_table.wl` packages these identities in `sFast[n,k]`. It first reduces k to its signed dyadic block, maps the local residue through the zero, plateau, reflection, and complement laws into the first quarter, and finally calls the finite diagonal formula. Thus an arbitrary entry can be evaluated without constructing the complete exponentially long block.

13.5 Block reduction in the diagonal coordinate

The direct sum (1.4) has only $\lfloor r/2 \rfloor + 1$ summands, but it is unsuitable when the reduced offset is very large. The block law of Chapter 11 first reduces an arbitrary offset modulo the block length, and a second idea removes the remaining dependence on the offset altogether.

The block law (11.5) first reduces an arbitrary m modulo $B = 2^{2n+1}$:

$$D_m(n) = \varepsilon_{\lfloor m/B \rfloor} D_{m \bmod B}(n). \quad (13.4)$$

Unlike the original code's block rule, this version acts directly on the natural diagonal offset. It avoids bookkeeping caused by the shift $n + 1$.

13.6 Logarithmic-time signed prefix moments

The direct sum is still unsuitable when the reduced offset is very large. The key observation is that, for fixed $n \geq 1$, its binomial kernel is a polynomial of degree $2n - 1$ in the Thue–Morse index q :

$$\binom{2n + m - 2q - 1}{m - 2q} = \binom{2n + m - 2q - 1}{2n - 1} = \sum_{d=0}^{2n-1} c_d(m, n) q^d. \quad (13.5)$$

Thus only finitely many signed prefix moments are required.

Definition 13.1 (Signed Thue–Morse prefix moments). For $N, d \in \mathbb{N}$, let

$$M_d(N) = \sum_{q=0}^{N-1} \varepsilon_q q^d. \quad (13.6)$$

The sum is exclusive at N ; $M_d(0) = 0$.

Theorem 13.2 (Binary divide-and-conquer recurrence). For $N, d \in \mathbb{N}$,

$$M_d(2N) = - \sum_{j=0}^{d-1} \binom{d}{j} 2^j M_j(N), \quad (13.7)$$

$$M_d(2N + 1) = M_d(2N) + \varepsilon_N (2N)^d. \quad (13.8)$$

The sum in (13.7) is empty when $d = 0$.

Proof. Pair the terms with indices $2q$ and $2q + 1$ and use (2.1):

$$\begin{aligned} M_d(2N) &= \sum_{q=0}^{N-1} \left(\varepsilon_{2q}(2q)^d + \varepsilon_{2q+1}(2q+1)^d \right) \\ &= \sum_{q=0}^{N-1} \varepsilon_q \left((2q)^d - (2q+1)^d \right) \\ &= - \sum_{j=0}^{d-1} \binom{d}{j} 2^j \sum_{q=0}^{N-1} \varepsilon_q q^j. \end{aligned}$$

The highest-degree terms cancel. Adding the final term at index $2N$ gives (13.8). $\square$

Let $L = \lfloor m/2 \rfloor + 1$. Combining (13.5) with (13.6) gives

$$D_m(n) = \sum_{d=0}^{2n-1} c_d(m, n) M_d(L) \quad (n \geq 1). \quad (13.9)$$

The recursion depth for $M_d(L)$ is $O(\log L)$. At each level, computing all moments through degree $2n - 1$ by (13.7) uses $O(n^2)$ arithmetic operations. Ignoring integer multiplication bit costs, the evaluator therefore uses

$$O(n^2 \log m) \quad (13.10)$$

exact arithmetic operations, rather than $O(m)$ summands.

Algorithm 13.3 (Hybrid evaluator). Given $n, k \in \mathbb{N}$:

1. If $k \leq n$, return 0.
2. Set $m = k - n - 1$, $B = 2^{2n+1}$, and write $m = qB + r$.
3. Compute the core value $D_r(n)$ by the direct sum if r is below a chosen small cutoff; otherwise use (13.9).
4. Return $\varepsilon_q D_r(n)$.

The cutoff only chooses between two exact algorithms; it does not affect the result. The supplied implementation uses 128 by default.

```
tmPrefixMoments[0, maxDegree_Integer?NonNegative] :=
  ConstantArray[0, maxDegree + 1];

tmPrefixMoments[length_Integer?Positive, maxDegree_Integer?NonNegative] :=
  tmPrefixMoments[length, maxDegree] = Module[
    {half = Quotient[length, 2], lower, paired},
    lower = tmPrefixMoments[half, maxDegree];
    paired = Table[
      If[d == 0, 0,
        -Sum[Binoial[d, j] 2^j lower[[j + 1]], {j, 0, d - 1}]],
      {d, 0, maxDegree}
    ];
    If[OddQ[length],
      paired += tmSign[half] Table[(2 half)^d, {d, 0, maxDegree}];
    ];
    paired
  ];
```

For $n \geq 1$, the polynomial coefficients $c_d(m, n)$ are obtained exactly from

$$\frac{1}{(2n-1)!} \prod_{j=1}^{2n-1} (m - 2q + j).$$

The full source file `diagonal_polynomials.wl` contains this contraction, block reduction, a compact replacement for the original `s`, and a literal repeated-prefix reference implementation.

13.7 Memory and implementation notes

The memoized moment routine stores vectors keyed by the pair $(N, d_{\max})$. A single query touches only the binary ancestors $N, \lfloor N/2 \rfloor, \lfloor N/4 \rfloor, \dots$, so its live recursion depth is logarithmic. For a long batch with many unrelated degree bounds, one may remove memoization or cache moments at a fixed maximum degree and truncate as needed.

Exact integer sizes can be very large. Complexity (13.10) counts arithmetic operations, not bit operations; this is the appropriate distinction because the output itself may contain thousands or millions of bits. The algorithm avoids artificial work proportional to the index m , but it cannot avoid the cost of storing the exact answer.

The local recurrence (3.8) remains preferable when a dense rectangular region is desired. The moment method is designed for sparse random access to isolated entries. The diagonal polynomial and Newton–Bell methods are designed for symbolic dependence on n . Keeping these use cases separate leads to simpler and faster code.

The exact theorems of Chapters 8 and 9 also translate directly into inexpensive integer tests; the following routines, from the third report’s implementation, are retained as `verification/source3/thue_morse_diagonals.wl`.

13.8 Root and denominator utilities

The exact theorems translate directly into inexpensive integer tests.

Listing 13.5: Exact rational-root tests and coefficient denominator.

```
ClearAll[halfGridRootIndices, halfGridFactor,
  negativeHalfGridValue, negativeHalfGridRootQ,
  minimalMonomialDenominator];

halfGridRootIndices[d_Integer?Positive] := Module[
  {q = d - 1},
  If[q == 0, {},
    Select[Range[0, q - 1],
      Mod[q, 2^(# + 1)] >= 2^(# + 1) - (# + 1) &]
  ]
];

halfGridFactor[d_Integer?Positive, var_: x] :=
  Times @@ (2 var - # & /@ halfGridRootIndices[d]);

(* The point tested here is x=-m/2, m>=1. *)
negativeHalfGridValue[d_Integer?Positive,
  m_Integer?Positive] := Module[{q = d - 1},
  Sum[(-1)^j Binomial[m - 1, j]
    If[q - j < 0, 0, tmSign[q - j]], {j, 0, m - 1}]
];
```

```

negativeHalfGridRootQ[d_Integer?Positive,
  m_Integer?Positive] :=
  PossibleZeroQ[negativeHalfGridValue[d, m]];

minimalMonomialDenominator[d_Integer?Positive] := Module[
  {q = d - 1, c, v2},
  c = Ceiling[q/2];
  v2 = IntegerExponent[Factorial[q], 2];
  Factorial[q]/2^Min[v2, c]
];

```

For instance,

```

Table[
  {d, halfGridRootIndices[d],
    Factor[halfGridFactor[d, x]],
    minimalMonomialDenominator[d]},
  {d, 1, 32}
]

```

returns the exact nonnegative rational factor and least coefficient denominator for each of the first thirty-two diagonals without factoring the full polynomial. Negative candidates are tested with `negativeHalfGridRootQ[d, m]`, which evaluates the polynomial at $-m/2$ by the finite-difference formula rather than by expansion.

13.9 Independent consistency checks

The supplied `thue_morse_diagonals.wl` file also contains a literal memoized prefix-sum implementation and the following checks:

```

verifyPromptDiagonals[]
verifyClosedFormula[5, 50]
verifyRecurrence[5, 50]

```

The first compares the general formula with all eight explicit polynomials of the originating program. The second compares the sparse formula, literal repeated prefix sums, and the signed block formula. The third verifies the weighted row recurrence [Equation \(1.2\)](#). These are regression tests; the preceding sections provide the proofs.

Chapter 14

Computational verification

The archive accompanying this volume contains `experiments.py`. Its identities are checked with Python integers and SymPy exact rationals; floating-point arithmetic is used only to draw Figure 11.1. The program performs the following independent comparisons.

1. It builds the table from the original weighted recurrence using prefix sums to evaluate each weighted convolution in constant time after preprocessing one row.
2. It evaluates $D_{k-n-1}(n)$ directly from (1.4) and compares every tested entry with the recurrence.
3. It constructs the finite block B_n by sliding-window convolutions and compares the signed-block lookup with both previous methods.
4. It mirrors the branch order of the Wolfram Language hybrid evaluator `sFast`, checks every residue of the tested row blocks, and then checks several complete signed repetitions.
5. It constructs the same diagonal polynomials independently from the ruler recurrence and verifies the half-step and differential identities symbolically.
6. It checks support, reflection, complement, plateau, total mass, multiplicities, and distinct-value counts for several rows and for arbitrary summation orders h through the tested range.
7. It verifies the nonnegative half-integer root criterion and the negative half-integer finite-difference formula, including the two valuation criteria (9.17)–(9.18).

A successful run prints

```
All exact checks passed.
```

and writes a text report of the initial factorizations. These experiments are regression checks rather than substitutes for the proofs above.

14.1 The three retained verification programs

Each source shipped an exact verification program together with a Wolfram Language implementation, and all three are retained unchanged under `verification/` in the package directory, with the outputs of the runs that accompanied the sources. Their checks overlap, and they were written independently, so agreement between them is evidence about the normalizations rather than about a shared implementation.

- `verification/source1/experiments.py` is the program described above, with `thue_morse_table.wl` (the hybrid evaluator `sFast`) and `verification_report.txt`.
- `verification/source2/diagonal_polynomials.py` verifies, in exact SymPy arithmetic, the direct and Newton–Bell generators, the half-step law, literal repeated prefix summation,

finite-block reduction, the prefix-moment evaluator of [Chapter 13](#), primitive normalization, exact denominators, and both half-grid formulas; run with `-max-degree 12 -verify`, and with `-scan-negative-half` for the finite scan reported after [Theorem 9.8](#) (no counterexample for odd $b \leq 63$, $m \leq 10000$). Its Wolfram Language companion `diagonal_polynomials.wl` contains the moment contraction, block reduction, a compact replacement for the original `s`, and a literal repeated-prefix reference implementation; `VERIFICATION.txt` records the commands, versions and results.

- `verification/source3/diagonal_analysis.py` performs the checks listed next, writes `generated/verification_report.txt`, `generated/diagonal_polynomials.csv` and `generated/half_grid_roots.csv`, and regenerates the normalized-pulse figure; `thue_morse_diagonals.wl` holds the routines of [Chapter 13](#) and the consistency checks `verifyPromptDiagonals`, `verifyClosedFormula` and `verifyRecurrence`.

14.2 Exact Python verification

The archive includes `diagonal_analysis.py`. It uses Python integers for the discrete checks and SymPy rational polynomials for symbolic expansion. Running

```
python diagonal_analysis.py --output-dir generated --max-table-q 20
```

creates:

- `verification_report.txt`,
- `diagonal_polynomials.csv`,
- `half_grid_roots.csv`, and
- the PDF and PNG versions of the normalized-pulse figure, retained as `figures/c_normalized_row_pulses.pdf`.

The bundled run performed the following exact checks.

1. The sparse formula, literal repeated prefix sums, and signed block formula agree for $0 \leq n \leq 5$, $0 \leq k \leq 70$.
2. The weighted recurrence agrees on the same range for $1 \leq n \leq 5$.
3. Polynomial evaluation agrees with the integer sparse sum for $0 \leq q \leq 30$.
4. The nonnegative half-grid zero criterion agrees for $0 \leq q, m \leq 30$.
5. The negative half-grid finite-difference formula agrees with exact polynomial evaluation for $0 \leq q \leq 30$, $1 \leq m \leq 16$.
6. The trailing-ones negative-root theorem agrees for exponents $2 \leq a \leq 9$ in four consecutive dyadic blocks.
7. The exact minimal denominator formula agrees with coefficient inspection for $0 \leq q \leq 24$.
8. Pulse support, positivity, palindromy, mass, maximum, and plateau length agree for prefix orders 1 through 10.

All assertions use exact equality. The numerical root quoted after [Theorem 9.2](#) is the only floating-point datum in the main mathematical discussion.

14.3 Selected continued factorizations

The polynomial law continues well beyond the eight hand-entered cases. The following table records exact rational linear factors found by symbolic factorization. The nonnegative factors are predicted a priori by [Theorem 9.2](#); negative factors illustrate the additional structure still to be classified.

Table 14.1: Selected exact rational factors of D_q .

d	q	rational linear factor product	residual degree
1	0	1	0
2	1	x	0
3	2	$(x+1)(2x-1)$	0
4	3	$x(x+2)(2x-1)$	0
5	4	1	4
6	5	$x(x-1)$	3
7	6	$(x-1)(x+1)(2x-1)$	3
8	7	$x(x-1)(x+2)(2x-1)$	3
9	8	$x+1$	7
10	9	$x(x+2)$	7
11	10	$(x+1)(x+3)(2x-1)$	7
12	11	$x(x+2)(x+4)(2x-1)$	7
13	12	$(x+5)(2x-3)$	10
14	13	$x(x-1)(x+6)(2x-3)$	9
15	14	$(x-1)(x+1)(x+7)(2x-3)(2x-1)$	9
16	15	$x(x-1)(x+2)(x+8)(2x-3)(2x-1)$	9

The factors in [Table 14.1](#) were verified by exact polynomial division. No irreducibility claim is made for the residual factors unless a computer algebra system has certified it over $\mathbb{Q}$; the table is intended to display recurring linear structure, not to replace the general theorems.

Chapter 15

Further deductions and research directions

15.1 Polynomial interpolation in a discrete parameter

The variable x is not merely a formal row coordinate. Under $h = 2x + 1$, it continuously interpolates the discrete order of summation. The half-step recurrence is exactly the identity that one additional summation in h produces one lower-index coefficient after taking a difference in the order parameter. This may be useful when comparing different repeated- integration normalizations in the Fabius corpus.

15.2 Root automata and residual factors

The nonnegative half-integer roots are completely classified by a terminal block condition. At a fixed negative half-integer, the zero test is a finite linear combination of a bounded window of Thue–Morse signs. This suggests a systematic finite-automaton study of the residual linear factors of D_r as r varies. The examples $x + 1$ and $x + 2$ already reduce to elementary 2-adic valuation parity. Higher backward differences produce richer but still finite digital conditions.

15.3 Other Mahler products

The deflation theorem (4.4) applies whenever a base generating function contains a factor $(1 - z)^c$ and then contracts to z^b . For products of the form

$$E_b(z) = \prod_{j \geq 0} (1 - z^{b^j}), \quad E_b(z) = (1 - z)E_b(z^b),$$

one obtains an analogous sparse diagonal polynomial with degree gaps in multiples of b . Finite-level factorization then produces blocks of length b^h . Positivity and the exact plateau geometry require additional care when the quotient factors are no longer all-one intervals, but the polynomial mechanism survives unchanged.

15.4 Scaling regimes beyond fixed diagonals

The fixed- m asymptotic (6.19) is only the extreme left edge. There are at least four natural regimes for $m = m(n)$:

1. m fixed;
2. $m \rightarrow \infty$ but $m = o(2^n)$, where only the first few geometric upper bounds in (11.15) are active;

3. m proportional to the standard deviation of the restricted-uniform sum, where a local central limit theorem should control $h_{2n,m}$;
4. m within $O(n)$ of the exact plateau and zero-run boundaries, where finite edge corrections dominate.

A uniform saddle-point analysis of $H_{2n}(z)$ could connect the polynomial edge law to the Gaussian bulk and the exact flat plateau without passing through separate approximations.

15.5 Residual factors and zero geometry

After removing the completely classified nonnegative rational factor R_m^+ and the negative integer factors detected by (9.19), the remaining factor often appears irreducible over $\mathbb{Q}$ in low degrees. Questions include:

- Are all nonnegative half-integer zeros simple?
- Can the negative integer roots be characterized by binary automata in the pair (m, r) from the binomial sums at $x = -r$?
- Does Theorem 9.8 hold?
- What is the limiting distribution of the nonrational complex zeros of D_m after an m -dependent scaling?

The half-step lowering law suggests that finite-difference analogues of classical results on Appell and Sheffer zero interlacing may be relevant, but $E(z^2)$ is not a positivity-preserving prefactor, so standard real-rooted arguments do not apply directly.

15.6 Automatic and p -adic structure

The cumulants (7.9) depend on r only through $\nu_2(r)$. This makes the Newton recurrence an unusual hybrid of Bell-polynomial and automatic-sequence structure. It is natural to ask for:

- finite automata or linear representations for coefficient residues of the primitive polynomials $\tilde{D}_m$ modulo odd primes;
- exact Newton polygons of $\tilde{D}_m$ at 2 beyond the coarse bound (8.7);
- analogues for generalized base- b Thue–Morse products $\prod_{j \geq 0} (1 - z^{b^j})$ or for weighted digit characters.

For a general digit character, the logarithmic cumulants are controlled by b -adic divisibility, so much of the Riordan and Bell machinery survives with ν_2 replaced by ν_b and with a different finite block.

15.7 Binary classification of negative rational factors

The nonnegative rational factors have the residue-window description Equation (9.3). On the negative side, Equation (9.16) gives a complete finite-difference test, but it is not yet a comparably transparent description in the binary digits of q and m . A natural next problem is to characterize the zero set

$$\{(q, m) \in \mathbb{N} \times \mathbb{N}_{>0} : \nabla^{m-1} \varepsilon_q = 0\} \quad (15.1)$$

in terms of carries, complemented suffixes, or a finite collection of binary recursions. The trailing-ones theorem Theorem 9.9 is one infinite stratum of this set.

Because ε_q is a two-automatic sequence, the bivariate zero mask in Equation (15.1) is a promising candidate for analysis by regular-sequence and linear-representation methods. Even a classification for fixed m , or for m restricted to powers of two, would sharpen the factor tables substantially.

15.8 The residual root geometry

Remove from D_q every rational linear factor detected by [Theorems 9.2 and 9.6](#). The remaining factor can have irrational real roots as well as complex-conjugate pairs. The following questions are concrete and computationally accessible.

1. How many real roots can the residual factor have, and how does this number depend on the binary expansion of q ?
2. Is there a useful vertical strip or parabolic region containing all roots after centering by the exact mean $-(q-1)/4$?
3. Do the empirical root measures have a limiting distribution after an appropriate scaling in q ?
4. Which diagonals have repeated roots? The generating and difference identities provide exact resultant recurrences with which to test square-freeness.

The exact root moments in [Equations \(6.16\) and \(6.18\)](#) give two nontrivial constraints on any proposed limiting law.

15.9 Joint asymptotics beyond a fixed diagonal

Formula [Equation \(6.19\)](#) treats q as fixed while $n \rightarrow \infty$. The Fabius scaling in [Theorem 12.2](#) instead has $q \asymp 2^{2n}$. Between them lie several unexplored regimes:

$$q = o(n), \quad q \asymp n, \quad q \asymp n^2, \dots, \quad q \asymp 2^{2n}. \quad (15.2)$$

The coefficient integral

$$D_q(n) = \frac{1}{2\pi i} \oint \frac{E(z^2)}{(1-z)^{2n}} \frac{dz}{z^{q+1}} \quad (15.3)$$

provides a common starting point. For subexponential q , the saddle comes mainly from the negative-binomial factor; as it approaches dyadic scales, zeros and oscillations of the lacunary factor $E(z^2)$ can no longer be treated as a perturbation. A uniform bridge between these regimes would connect the polynomial theory directly to the sharp Fabius asymptotics.

15.10 Fast coefficient queries inside one pulse

Constructing all coefficients of Q_r costs $O(2^r)$ memory, although the sliding-window recurrence is optimal when the whole pulse is required. For one coefficient $a_{r,b}$, it should be possible to exploit the binary expansion of b , palindromy, plateau structure, and recursive interval convolution to obtain substantially smaller memory use. Useful targets include:

- a divide-and-conquer exact evaluator for one coefficient;
- certified modular evaluation followed by Chinese remaindering;
- asymptotically sharp local-limit estimates with rigorous error terms;
- an adaptive method that switches between the sparse diagonal sum and the dyadic pulse representation according to $q = k - n - 1$.

15.11 Formalization targets

The repository already crosswalks Lean proofs of the exact prefix formula, iterated-prefix convolution, finite dyadic quotient, terminal zero run, signed reversal, finite approximation polynomials, corrected

normalization and step approximants [1, 4, 5]. The statements of this volume reduce to finite sums, polynomial identities, formal power-series coefficient extraction, and elementary integer divisibility once those inputs are imported. The natural additions, in dependency order, are:

1. the shifted-table identity (3.4) and the local discrete equation (3.8);
2. a polynomial-valued definition of D_r , the generating function (7.2), the sparse formula (1.4), and the evaluation theorem $s_{n,n+1+r} = D_r(n)$;
3. the half-step, ruler and differential recurrences and the diagonal Mahler equation (7.19);
4. the finite-block theorem (10.11) with the plateau, complement, multiplicity and distinct-value statements for Q_h and B_n ;
5. the half-grid transposition (9.1), the exact half-integer zero criterion, and the negative half-lattice formula;
6. primitive normalization, the exact denominator theorem, and the rational-root restriction;
7. the trailing-ones negative-root family (9.21);
8. the signed-prefix-moment recurrences and the correctness of the fast evaluator;
9. the exact row-histogram identification (12.9) against the existing approximation modules.

None of these is claimed to have been formalized; the list is a technical implementation guide.

Appendix A

A compact theorem dictionary

For reference, the principal objects and translations are collected here.

Object	Exact formula or relation
Signed Thue–Morse	$\varepsilon_q = (-1)^{s_2(q)}$
Lacunary series	$E(z) = \sum_q \varepsilon_q z^q = \prod_{j \geq 0} (1 - z^{2^j})$
Mahler equation	$E(z) = (1 - z)E(z^2)$
Exclusive prefix	$S(2q) = 0, S(2q + 1) = \varepsilon_q$
Iterated prefix	$\sum_r T_r^{(h)} z^r = E(z)/(1 - z)^h$
Row generating function	$A_n(z) = z^{n+1} E(z)/(1 - z)^{2n+1}$
Row/prefix map	$s_{n,n+1+r} = T_r^{(2n+1)}$
Diagonal polynomial	$D_r(x) = [z^r] E(z^2)/(1 - z)^{2x}$
Original diagonal d	$P_d(x) = D_{d-1}(x)$
Half-step	$D_r(x) - D_r(x - 1/2) = D_{r-1}(x)$
Ruler coefficient	$\lambda_m(x) = 2x + 2 - 2^{\nu_2(m)+1}$
Ruler recurrence	$rD_r = \sum_{m=1}^r c_m D_{r-m}$
Row block length	$L_n = 2^{2n+1}$
Row maximum	$M_n = 2^{\binom{2n}{2}}$
Signed block law	$s_{n,qL_n+a} = \varepsilon_q b_{n,a}$

Appendix B

Nonnegative rational factors through $m = 31$

The following table lists, for each m , the set $\mathcal{A}_m$ of those $a \geq 0$ with $m \bmod 2^{a+1} \geq 2^{a+1} - a - 1$, so that $\prod_{a \in \mathcal{A}_m} (2x - a)$ is the systematic factor Φ_m of (9.7). The displayed squarefree product $\prod_{a \in \mathcal{A}_m} (2x - a)$ contains one copy of every nonnegative rational linear factor; an empty set means that D_m has no nonnegative rational zero.

m	$\mathcal{A}_m$	m	$\mathcal{A}_m$
0	$\emptyset$	16	$\emptyset$
1	$\{0\}$	17	$\{0\}$
2	$\{1\}$	18	$\{1\}$
3	$\{0, 1\}$	19	$\{0, 1\}$
4	$\emptyset$	20	$\emptyset$
5	$\{0, 2\}$	21	$\{0, 2\}$
6	$\{1, 2\}$	22	$\{1, 2\}$
7	$\{0, 1, 2\}$	23	$\{0, 1, 2\}$
8	$\emptyset$	24	$\emptyset$
9	$\{0\}$	25	$\{0\}$
10	$\{1\}$	26	$\{1\}$
11	$\{0, 1\}$	27	$\{0, 1, 4\}$
12	$\{3\}$	28	$\{3, 4\}$
13	$\{0, 2, 3\}$	29	$\{0, 2, 3, 4\}$
14	$\{1, 2, 3\}$	30	$\{1, 2, 3, 4\}$
15	$\{0, 1, 2, 3\}$	31	$\{0, 1, 2, 3, 4\}$

The table makes the nested terminal windows before powers of two visible. At $m = 2^{a+1} - a - 1, \dots, 2^{a+1} - 1$, the new root $x = a/2$ appears; lower-period root patterns continue independently underneath it.

Appendix C

Formula sheet for implementation

For convenient reference, the principal formulas are collected here.

quantity	exact formula
signed Thue–Morse value	$\varepsilon_j = (-1)^{s_2(j)}$
order- r prefix series	$\sum_{q \geq 0} T_q^{(r)} z^q = E(z)/(1-z)^r$
whole table	$s_{n,k} = T_{k-n-1}^{(2n+1)}$
row series	$A_n(z) = z^{n+1} E(z)/(1-z)^{2n+1}$
full bivariate series	$\sum s_{n,k} u^n z^k = z(1-z)E(z)/((1-z)^2 - uz)$
positive pulse	$Q_r(z) = \prod_{j=1}^{r-1} (1+z+\dots+z^{2^j-1})$
block law	$T_{a2^r+b}^{(r)} = \varepsilon_a[z^b]Q_r(z)$
terminal zero test	$T_{a2^r+b}^{(r)} = 0 \iff b \geq 2^r - r$
diagonal polynomial	$D_q(x) = \sum_{j \leq q/2} \varepsilon_j (2x)^{q-2j} / (q-2j)!$
diagonal generating function	$\sum_{q \geq 0} D_q(x) z^q = E(z^2)/(1-z)^{2x}$
direct table evaluator	$s_{n,k} = \sum_{j \leq (k-n-1)/2} \varepsilon_j \binom{n+k-2j-2}{k-n-1-2j}$ for $k > n$
minimal coefficient denominator	$q! / 2^{\min(\nu_2(q!), \lceil q/2 \rceil)}$
nonnegative rational root	$m/2$ is a root iff $q \bmod 2^{m+1} \geq 2^{m+1} - (m+1)$
negative rational root	$-m/2$ is a root iff $\sum_{j=0}^{m-1} (-1)^j \binom{m-1}{j} \varepsilon_{q-j} = 0$
Fabius sample	$s_{n,n+1+\lfloor 2^{2n}x \rfloor} / 2^{\binom{2n}{2}} \rightarrow F(x)$

Appendix D

Source provenance, notation dictionary, and reproducibility

D.1 Sources and consolidation

This volume consolidates the three diagonal-polynomial reports filed in the repository’s frontier intake of September 3, 2026, under `drafts/thue-morse/`: `thue_morse_diagonal_polynomials` (*Diagonal Polynomials and Dyadic Block Geometry in Repeated Thue–Morse Summation*, the spine of this volume), `thue_morse_diagonal_polynomials-2` (*Diagonal Polynomial Laws in Odd Iterated Thue–Morse Summation: Riordan-array structure, 2-adic Bell recurrences, exact arithmetic, and fast Wolfram Language evaluation*), and `thue_morse_diagonal_polynomials_article_and_code` (*Diagonal Polynomials and Dyadic Block Geometry in Repeated Thue–Morse Prefix Summation: Exact formulas, denominator laws, rational roots, and fast Wolfram Language evaluation*). Their $\text{\LaTeX}$ sources, PDFs and READMEs were removed when this volume was filed; the repository history retains them. The verification programs, Wolfram Language files, generated data, and figures were retained unchanged.

The first report supplies the chapter structure and the statements and proofs of every result all three shared. The second report contributed the local discrete equation, the Riordan-array chapter with its general subdiagonal theorem, the binomial-basis integer coefficients, the partition and determinant forms of the Bell representation, the half-grid translation formulas and the integer-step relation, the arithmetic chapter (primitive normalization, exact denominator, rational-root restriction), the parity-split negative half-grid formula and the conjecture on strict negative half-integers, the concrete block examples and table, the Lean-level identification of the block polynomial with the approximation polynomial, the logarithmic-time prefix-moment evaluator, the scaling regimes and automatic-structure questions, and the factor table through $m = 31$. The third report contributed the first-rows table, the first algebraic moments of the roots and the root product, the fixed-diagonal asymptotic form, the diagonal Mahler equation, the half-grid transposition theorem, the complete description of the nonnegative rational roots with the counting bound and the Mersenne chain, the irrational-root example, the trailing-ones family of negative integer roots, the root and denominator utilities and consistency checks, the continued-factorization table, the exact row–histogram identification and the table-limit theorem, the negative-factor classification and residual-geometry questions, and the implementation formula sheet.

Before deduplicating, the following identifications were checked: the three table identifications and row generating functions agree; the three sparse formulas, monomial-coefficient formulas and top-three layers agree, and the root moments of the third report follow from them; the block polynomials Q_h , H_{h-1} and $\mathcal{A}_h$ are the same product, with the same mass $2^{\binom{h}{2}}$, maximum $2^{\binom{h-1}{2}}$ and plateau; the half-integer criteria agree; the negative-root criteria of the first and second reports agree after the substitution $r = 2q$ or $r = 2q + 1$, and the trailing-ones family of the third is consistent with both; the exact denominators 1, 1, 1, 3, 6, 15, 90, 315, 2520, $\dots$ agree between the second and third reports. No

discrepancy was found.

D.2 Notation dictionary

This volume	Second report	Third report	Meaning
$E(z)$	$\mathcal{T}(z)$	$\mathcal{K}(z)$	$\sum \varepsilon_q z^q = \prod (1 - z^{2^j})$
$s_{n,k}$	$s(n, k)$	$s_{n,k}$	the table
$T_r^{(h)}$	$S_r^{(h)}, \mathcal{P}^h \varepsilon$	$\sigma_h(r)$	h -fold inclusive prefix
$A_n(z)$	$G_n(z)$	$T_n(z)$	row generating function
$D_r(x)$	$D_m(x)$	$D_q(x)$	the diagonal polynomial
$G_x(z)$	$F(x, z)$	$F_x(z)$	$\sum_r D_r(x) z^r$
$\widehat{D}_r = r! D_r$	P_m	$\widehat{D}_q, E_q$	the Sheffer-normalized polynomial
$\widetilde{D}_r = r! D_r / 2^{\lceil r/2 \rceil}$	Q_m	H_q	the primitive normalization
$\lambda_m(x)$	$\lambda_r(x)$	—	ruler cumulants $2x + 2 - 2^{\nu_2+1}$
$Q_h(z), q_{h,a}$	$H_{h-1}(z), h_{h-1,r}$	$\mathcal{A}_h(z), a_{h,b}$	the block polynomial of order h
$L = 2^h, M_h$	B_{h-1}, C_{h-1}	$2^h, 2^{\binom{h-1}{2}}$	block length and maximum
$B_n(z), b_{n,a}, L_n, M_n$	$c_{n,r}, B, C$	B_n, H_n, M_n	the row block
$s_2(q)$	$s_2(q), \text{popcount}$	$\text{wt}_2(q)$	binary digit sum

D.3 Retained files

- `verification/source1/`: `experiments.py`, `thue_morse_table.wl`, `verification_report.txt`, `row_profiles.pdf`, `row_profiles.png`.
- `verification/source2/`: `diagonal_polynomials.py`, `diagonal_polynomials.wl`, `VERIFICATION.txt`.
- `verification/source3/`: `diagonal_analysis.py`, `thue_morse_diagonals.wl`, `generated/` (verification report, run transcript, the two CSV tables, the figure).
- `figures/`: the two profile figures used in the text.

The complete Wolfram Language and Python sources that the second report printed as appendices are not reproduced in the text; they are the retained files. The volume loads `docs/fabius-notation.tex` by relative path; with a $\text{\TeX}$ installation containing Libertinus (or, as a fallback, Latin Modern) and the listed standard packages, compile with three passes of

```
pdflatex -interaction=nonstopmode -halt-on-error \
  Thue_Morse_Diagonal_Laws.tex
```

No Lean files were added or compiled for this volume.

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